

Reward Poisoning at Runtime-Assurance Authority Transitions

Abstract—Runtime assurance can mask an unsafe raw policy while a trusted mechanism retains action authority during adaptation. Retiring that authority can therefore expose a policy trained from update evidence whose provenance was not re-established for raw release. We examine this authority-transition vulnerability under bounded reward-record poisoning. In a third-party shield-retirement lifecycle, poisoning increased raw-policy violation rates at retirement by factors of 3.6 and 1.6 in two of three provenance-complete paired REINFORCE training runs. A freshly regenerated third pair was unchanged, and protected execution remained violation-free. A controlled Safe-Control-Gym study isolated the same mechanism: poisoned raw release failed on 27/120 evaluated snapshot–state pairs, compared with 2/120 for paired clean release and 0/120 under resident predictive authority. Across a locked 50-run obstacle corpus, all 11 positive outcomes occurred below 0.06 clean raw-policy/shield disagreement, whereas none of 32 high-disagreement runs was positive. Seven of 18 low-disagreement runs were non-positive; the separate fresh third-party pair was also low-disagreement and non-positive. Retirement-state disagreement thus identifies an opportunity region, not an attack-success predictor. Provenance binding and trusted recomputation close the upstream channel; commit admission and retained authority contain its downstream consequence.

I. INTRODUCTION

Runtime assurance can keep an adaptive controller physically safe without establishing that its raw policy is safe. A resident decision module may replace unsafe policy proposals with actions from a trusted baseline while corrupted update data continue to influence learning. The assured composition remains protected, but the raw policy can accumulate latent damage. A security boundary appears when the decision module is retired and the raw policy inherits action authority.

Existing assurance and learning-security mechanisms address complementary parts of this lifecycle. Simplex retains a decision module and an assured baseline controller [1]. Neural and barrier-based Simplex extend this model with online retraining and stronger switching conditions [2], [3]. Policy repair and safe projection constrain learned policies before deployment [4]–[6]. Reward-poisoning work, by contrast, studies how adversarial learning signals redirect policy updates. The unresolved composition is an authority transition in which protected adaptation uses an adversarial update log, followed

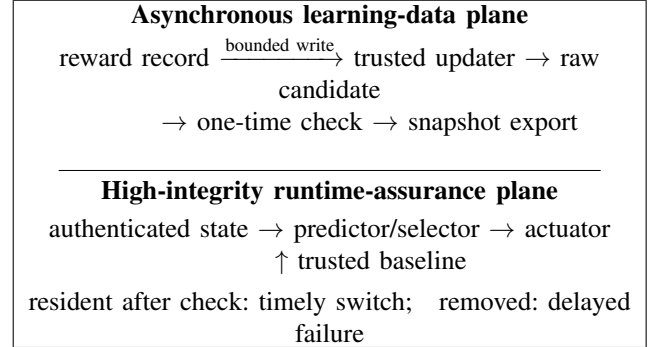


Fig. 1. Evaluated trust boundary. The benchmark begins after bounded reward-record access is obtained; it does not model the initial service compromise. Upstream provenance or recomputation removes this attack path, whereas resident predictive authority contains its downstream physical consequence. The same boundary is evaluated on a third-party shield-retirement workflow in Section VII.

by deployment of the raw policy without the mechanism that governed data collection.

We examine this composition under an explicit split-trust precondition. Operational technology (OT) guidance separates runtime-control assets from data-management services and historians that exchange information across zones [7], [8]. Our threat model begins after compromise of a reward-ingestion principal in this learning-data plane. The attacker can modify bounded scalar reward records consumed by a policy-gradient updater. Authenticated runtime states, actions, learner code, parameters, optimizer state, certificates, and deployment states remain outside its write authority. Trusted recomputation from authenticated transitions blocks the evaluated channel when reward semantics permit it.

The attack targets the evidence used to authorize raw release. Bounded reward edits steer the raw policy toward a target that passes a finite release check but can fail later after runtime authority is removed. We call the resulting update *certificate-evasive* with respect to that finite release contract. The term does not imply evasion of reward-integrity checks or sound invariant certificates.

For a fixed policy-gradient batch, bounded reward edits induce a structured set of possible parameter updates. Advantage standardization makes this set non-affine, so we derive its normalized reward-influence geometry. Conic homogenization then yields exact positive support against linear release-gate halfspaces under global gradient-norm clipping. This analysis characterizes batch-local influence; the evaluated multi-batch attack remains a fixed, bounded target-shaping rule.

A. Evidence Stack and Contributions

The paper combines an independently designed lifecycle with a controlled mechanism study. Carr et al.'s published workflow trains a policy behind a runtime shield and later retires the shield's authority [9]. We add bounded reward-record poisoning only during the protected bootstrap. In two of three provenance-complete paired REINFORCE training runs, poisoning increased raw-policy violations at retirement by factors of 3.6 and 1.6. A freshly regenerated third pair was unchanged despite low clean disagreement. The controlled Safe-Control-Gym study isolated the same transition and compared one-time raw release with resident predictive authority, commit admission, and permanent filtering. Together, the studies support an architectural vulnerability with conditional exploitability.

Our contributions are:

- We formalize an authority-transition vulnerability through an evidence tuple covering subject, authority, provenance, information, and horizon. Deployment must re-establish each component changed by the transition.
- We instantiate this vulnerability in Carr et al.'s third-party shield-retirement lifecycle. The paired positive and non-positive training realizations show both the exposure and the conditional nature of the fixed attack.
- We derive exact batch-local reward influence on linear release-gate margins under standardized advantages and global gradient-norm clipping. Controlled experiments then separate permanent raw release from resident predictive authority and one-step filtering.
- We characterize the empirical boundary of the attack across systems, learner implementations, integrity controls, and deployment contracts. High retirement-state disagreement excludes every observed positive outcome in the locked obstacle corpus, while low disagreement defines an opportunity region without guaranteeing success.

II. MODEL AND PROBLEM STATEMENT

A. Attacked Histories and Plausible States

We consider a discrete-time cyber-physical system

$$x_{t+1} = f(x_t, u_t, w_t), \quad \tilde{y}_t = h(x_t) + a_t, \quad (1)$$

where x_t is the physical state, u_t is the control input, and $w_t \in \mathcal{W}$ is ordinary process uncertainty. The term a_t denotes a false-data injection (FDI) signal applied to the measurement channel. The controller observes the attacked history

$$h_t = (\tilde{y}_{0:t}, u_{0:t-1}, \tilde{\ell}_{0:t}), \quad (2)$$

where $\tilde{\ell}_t = \ell_t + b_t$ denotes a learning signal modified by update-data corruption b_t . The general model includes rewards, logs, and auxiliary learning signals. The end-to-end experiments specialize this channel to bounded scalar reward-record edits. Sensor FDI creates uncertainty about the physical state, whereas update-data corruption can redirect learning

with trusted state observations. Either channel can change the action map produced from h_t .

Following severe sensor-attack safety work [10], an attacked history may correspond to multiple physical states. For an attack family $\mathcal{A}$, define the plausible-state set

$$X_{\mathcal{A}}(h_t) = \{x : \exists E \in \mathcal{A}, H_t(E) = h_t, x_t(E) = x\}. \quad (3)$$

This set defines the state semantics of a certificate under strategic FDI. A certificate evaluated only at a nominal estimate $\hat{x}(h_t)$ does not cover another compatible state in $X_{\mathcal{A}}(h_t)$.

B. Online-Updated Controllers

The controller is a history-dependent policy π_{θ_t} with parameters θ_t . It selects $u_t = \pi_{\theta_t}(h_t)$ and updates according to

$$\theta_{t+1} = U(\theta_t, h_t). \quad (4)$$

An update is *nontrivial* when it changes the controller's action map on a reachable or admissible history. Parameter changes that preserve the action map are behaviorally trivial under this definition.

C. Certificates and Certified Kernels

Let Γ denote an invariant-set, barrier, reachability, or runtime-shield certificate. At history h , the certificate induces actions that are valid for every plausible state:

$$K_{\Gamma}(h) = \{u \in \mathcal{U} : u \text{ satisfies } \Gamma \text{ for all } x \in X_{\mathcal{A}}(h)\}. \quad (5)$$

We call $K_{\Gamma}(h)$ the *certified kernel*. Severe-attack safety filters construct or act within such kernels for fixed controllers. Online learning additionally requires admission of the updated action map produced by U .

Definition 1 (Certified online learning). *A mechanism is a certified online learner at history h if it accepts a certificate for $\pi_{\theta_{t+1}}$, where $\theta_{t+1} = U(\theta_t, h)$. The accepted certificate asserts that the next certified action is sound for every state in $X_{\mathcal{A}}(h)$.*

Definition 2 (Learning-freeze frontier). *For a family of histories parameterized by attack strength or ambiguity radius, the learning-freeze frontier is the boundary at which $K_{\Gamma}(h)$ becomes empty, the candidate updated action leaves $K_{\Gamma}(h)$, or the candidate parameter leaves the corresponding certified set $\Theta_{\Gamma}(h)$. Beyond this boundary, a sound certified learner must reject the certificate, freeze the update, or obtain extra trusted state information.*

III. THREAT MODEL

A. System Roles and Deployment Contracts

The defender deploys an online-updated controller π_{θ_t} and a certificate mechanism. A severe-attack-aware estimator or filter can return a plausible-state set $X_{\mathcal{A}}(h_t)$ and certified action kernel $K_{\Gamma}(h_t)$. The defender may also possess a trusted state anchor, backup controller, or external fail-safe, but each is an explicit architectural assumption and cost.

TABLE I
EVALUATED WRITE BOUNDARY. “TRUSTED” MEANS OUTSIDE ATTACKER
WRITE AUTHORITY, NOT FORMALLY VERIFIED.

Object	Defender assumption	Attacker access
Runtime state, action, plant	Authenticated control path	No write
Baseline, selector, certificate	Trusted runtime code/state	No write
Transitions and learner code	Trusted after collection	Read only
Reward record at ingestion	No origin/semantic check	Bounded write scalar
Parameters and optimizer state	Written only by updater	No write
Held-out states and release rule	Fixed, honest evaluation	No write or selection

We distinguish two deployment families. Under *raw-policy deployment*, a safety layer protects adaptation before a raw policy snapshot $\pi_{\theta+}$ receives action authority. Under *permanently filtered deployment*, a sound filter F remains online and certifies the composition $F \circ \pi_{\theta+}$.

The evaluation instantiates four contracts within these families. *Permanent raw release* removes runtime authority after a one-time five-step check. *Commit admission* certifies a candidate snapshot on a declared state set and horizon before raw release. *Resident predictive authority* repeats the five-step check online and can transfer control to the trusted baseline. The *one-step resident kernel* applies a current-state action filter and serves as a negative control. Each name denotes a distinct evidence scope and authority contract.

Permanent raw release is an intentional handoff. Examples include protected commissioning followed by snapshot export and promotion to a lower-latency control target without the training-time predictor. The handoff avoids permanent computation, intervention, or dependence on a backup service. Section VIII measures these tradeoffs. Resident authority instead contains raw-policy failure within the assured composition.

B. Split-Trust Learning Pipeline

The least-privilege model separates a high-integrity runtime-assurance plane from an asynchronous learning-data plane. The runtime plane owns the authenticated plant state, action selector, actuator interface, trusted baseline, and certificate outcome. The learning plane consumes recorded transitions and scalar rewards through telemetry, a historian, a replay buffer, or an external key-performance-indicator interface. OT guidance recognizes this separation and the exchange of historian data across zones [7], [8]. The benchmark models the resulting interface without assuming a particular product.

The evaluated attack begins after compromise of a principal with bounded reward-field write access. This access permits post-computation, pre-update mutation by a replay writer or broker. A compromised reward producer is equivalent when it has the same bounded output authority. In that case, source authentication establishes origin but not reward semantics.

C. Attack Channels and Capabilities

The general history model permits two attack channels. Observation FDI a_t changes the state semantics of the history and enlarges $X_{\mathcal{A}}(h_t)$. Update-data corruption b_t modifies learning signals and can redirect $U(\theta_t, h_t)$ with trusted state observations. Either channel can alter the next action map.

The end-to-end learner experiments use the narrower reward-record channel. The white-box, batch-aware attacker observes the current learner and adaptation batch, but not the held-out deployment-state choice. It modifies only scalar rewards before an ordinary policy-gradient update, subject to a declared per-step L_{∞} budget. It cannot write actuator commands, physical states, observations, dynamics, gradients, policy parameters, optimizer state, deployment states, or certificate outcomes. The adaptation selector observes each proposed action before the plant step and retains physical authority during data collection.

D. Evidence Transfer Across Authority Transitions

The contracts above differ in their evidence obligations. We represent the evidence used by an assurance decision as

$$E = (s, a, p, i, \tau), \quad (6)$$

The *subject* s is the raw policy π_{θ} or the assured composition $F \circ \pi_{\theta}$. The *authority* a identifies the resident module that constrains applied actions, or its absence. The *provenance* p records whether learning signals came from a trusted or adversarial update process. The *information* i describes the state, observation, belief, or filtered action available at decision time. The *horizon* τ specifies the interval over which the evidence bounds behavior.

A contract transition $C_{\text{train}} \rightarrow C_{\text{deploy}}$ may change one or more tuple components. Evidence transfers only after every changed component is re-established under C_{deploy} :

- *subject*: evidence about the composition does not bound the raw policy, and a raw-policy audit does not certify the composition;
- *authority*: evidence collected while a resident module masked unsafe proposals describes the masked process, not the raw policy after the module is removed;
- *provenance*: a policy trained on an adversarial update log is not the output of the trusted process the evidence was computed on;
- *information*: behavior under the authority’s information set does not determine behavior under the deployment observation;
- *horizon*: a finite evaluation pass does not bound an unbounded future.

Proposition 1 (Transfer obligation). *Let E be evidence collected under contract C_{train} , and let the transition $C_{\text{train}} \rightarrow C_{\text{deploy}}$ change a nonempty subset of the components of the evidence tuple. Acceptance under C_{deploy} is unsound unless each changed component’s evidence obligation is re-established under C_{deploy} .*

Proof. Each component is a premise of the assurance decision. A conclusion derived from one premise set is not preserved after a premise is removed or weakened. In the evaluated permanent-release transition, the subject changes from the composition to the raw policy and resident authority is removed. The update provenance becomes adversarial, the policy acts on deployment observations, and a finite check authorizes an unmonitored future. The original evidence therefore cannot be reused without re-establishing these obligations. $\square$

The proposition supplies the security model used throughout the paper. Independent transfer evidence re-establishes obligations before authority removal [11]. Resident Simplex mechanisms preserve the authority component [1]–[3]. Carr et al.’s lifecycle supplies a published transition from shielded bootstrap to raw-policy authority [9]. Our case study adds adversarial update provenance to that transition, and the controlled study isolates the resulting mechanism.

E. Integrity Assumptions and Escape Conditions

The vulnerable configuration supplies neither origin integrity for the reward record nor trusted reward recomputation. Binding a reward, transition, action, producer identity, and time to an authenticated append-only record blocks post-ingestion mutation. If reward is a deterministic function of authenticated transitions, recomputation by a trusted principal completely blocks the evaluated channel. If reward depends on delayed quality labels, operator feedback, or an external KPI unavailable to the runtime plane, the system instead needs a trusted producer, redundant validation, or a detector; a signature applied to an already malicious value is insufficient.

These controls act at different boundaries. Provenance and recomputation remove the upstream attack precondition. A task-aware detector can reject suspicious batches, subject to its distributional and semantic assumptions. Commit admission restricts which learned snapshot can be released. Resident predictive authority contains the physical consequence after deployment. In the frozen audit, trusted recomputation achieved true-/false-positive rates (TPR/FPR) of 1.000/0. Simple task-aware checks achieved TPR above 0.90 and FPR at most 0.036, making the effective attack readily detectable under those assumptions.

F. Security and Availability Goals

We separate four attacker or failure objectives:

- an *unsafe certified action*, where the mechanism accepts an applied input that is unsafe for some state in $X_A(h_t)$;
- a *stale certified policy*, where the current applied input is filtered but the accepted raw updated action map leaves the kernel;
- *uncertified learning*, where a nontrivial update continues after certificate rejection; and
- *forced freeze*, where the attacker expands the plausible set or redirects the update until sound learning must stop or re-anchor.

The defender also measures availability through admitted state coverage, accepted update fraction, task reward, filter

intervention, rejection, certificate latency, and trusted-anchor use. Useful learning requires paired task improvement over always-freeze in addition to physical safety.

G. Scope Boundaries

Severe-attack filters, secure estimators, and fault-tolerant control-barrier function (CBF) methods can preserve safety when their assumptions keep $K_\Gamma(h_t)$ nonempty [10], [12], [13]. This study examines certificate coverage for learner updates and the costs of sound deployment contracts. Its empirical claims concern sampled finite-horizon certificates, not continuous-state invariance or population-wide learner-family generality. The evaluation begins after bounded reward-record access and excludes service exploitation, reward-monitor evasion, and compromise of trusted recomputation.

IV. ASSURANCE-CONTRACT FAILURE MODES

This section separates four failure modes that arise at different points in the assurance lifecycle.

Nominal-state certification. A nominal certificate evaluates an estimate $\hat{x}(h)$ or a reported sensor value. Under FDI, an accepted action may be safe for $\hat{x}(h)$ but unsafe for another state in $X_A(h)$. Severe-attack safety filters address this state-semantics failure.

Action-only filtering. A robust safety filter can project the current action into $K_\Gamma(h)$ when the kernel is nonempty. This operation protects the applied action at that step. The learner update remains a separate object. If $\pi_{\theta_{t+1}}(h)$ leaves $K_\Gamma(h)$, the certificate covers the filtered action but not the updated action map. Certified learning must therefore admit the updated policy as well as the current action.

Uncertified learning after rejection. When $K_\Gamma(h) = \emptyset$, a severe-attack filter can reject the action or enter fail-safe mode. An update computed from the same attacked history then falls outside the accepted certificate lifecycle. A fail-safe can protect the plant while learning is frozen, rolled back, or re-anchored.

Protected adaptation, unsafe deployment. An adaptation shield can keep every data-collection transition safe while corrupted reward records influence the learner. The trusted update rule can then produce an unsafe raw policy without a physical failure during adaptation. Releasing that policy after shield retirement exposes the delayed effect. Raw release therefore requires evidence for the policy that receives action authority.

These failure modes require separate metrics. An *unsafe certified action* counts accepted certificates whose applied action is unsafe for some plausible state. A *stale certified policy* counts accepted certificates for which the updated raw policy leaves the kernel. *Uncertified learning* counts nontrivial updates after certificate rejection. End-to-end studies add delayed deployment violations, admission coverage, paired clean/poison outcomes, update fraction, intervention, reward, and certificate runtime.

V. CONTRACT-AWARE UPDATE ANALYSIS

This section develops the update conditions used by the empirical studies. The current- and future-history gates distinguish action filtering from certified learning. The reward-influence analysis then characterizes how a bounded reward edit can change a linear release-gate margin within one policy-gradient batch. The evaluated multi-batch attack remains the fixed target-shaping rule described in Section VIII.

A. Current- and Future-History Gates

The certificate condition has two levels. The current-history condition requires the updated action at the observed history to remain certified. The lifecycle condition extends this requirement to future attacked histories reachable after the update.

Lemma 1 (Current-history gate). *Let $M = (\pi_\theta, U, \text{Cert})$ be a history-dependent online controller. For an attacked history h , let $X_{\mathcal{A}}(h)$ be the plausible-state set and $K_\Gamma(h)$ its certified kernel. If M accepts a sound certificate for $\pi_{\theta_{t+1}}$, where $\theta_{t+1} = U(\theta_t, h)$, then*

$$\pi_{\theta_{t+1}}(h) \in K_\Gamma(h). \quad (7)$$

Consequently, if $K_\Gamma(h) = \emptyset$, then M cannot both accept the certificate and allow a nontrivial online update at h .

Proof. The certified action must satisfy the certificate for every state compatible with the attacked history. These states form $X_{\mathcal{A}}(h)$, and their common certified actions form $K_\Gamma(h)$. Continued certified operation therefore requires $\pi_{\theta_{t+1}}(h) \in K_\Gamma(h)$. An empty kernel requires rejection, freeze, rollback, or trusted information that changes the plausible-state set. $\square$

Proposition 2 (Action filtering is not update certification). *Let $F(v, h)$ be an action filter that returns an applied input in $K_\Gamma(h)$ whenever the kernel is nonempty. Consider a mechanism that first computes $\theta^+ = U(\theta_t, h)$ and then applies*

$$u_t = F(\pi_{\theta^+}(h), h), \quad (8)$$

The mechanism accepts the update without constraining the raw updated policy π_{θ^+} . If $\pi_{\theta^+}(h) \notin K_\Gamma(h)$, the applied input can be certified while the updated learner remains uncertified. Action-only filtering therefore does not imply Lemma 1.

Proof. The filter property gives $u_t \in K_\Gamma(h)$, so the current actuator input can satisfy the certificate. Lemma 1 instead constrains the updated controller through $\pi_{\theta^+}(h) \in K_\Gamma(h)$. When this membership fails, the certificate covers only the filtered actuator channel. Accepting the update under that certificate creates a stale certified policy. $\square$

The lemma and proposition define local gates. A policy update also determines actions at future histories. Let $\mathcal{H}_{\mathcal{A}}(h, \theta^+)$ denote the attacked histories reachable after h under π_{θ^+} and attack family $\mathcal{A}$. Define the certified parameter set

$$\Theta_\Gamma(h) = \{\theta : \forall h' \in \mathcal{H}_{\mathcal{A}}(h, \theta), \pi_\theta(h') \in K_\Gamma(h')\}. \quad (9)$$

Theorem 1 (Future-history lifecycle gate). *Let $\theta^+ = U(\theta_t, h)$ be a candidate online update. Suppose M accepts a sound*

lifecycle certificate for π_{θ^+} from history h over attack family $\mathcal{A}$. Then

$$\theta^+ \in \Theta_\Gamma(h), \quad (10)$$

equivalently,

$$\forall h' \in \mathcal{H}_{\mathcal{A}}(h, \theta^+), \quad \pi_{\theta^+}(h') \in K_\Gamma(h'). \quad (11)$$

If membership fails for a reachable h' , accepting the update creates a stale certified policy. If $K_\Gamma(h') = \emptyset$, certified learning must reject, freeze, roll back, or obtain trusted information that changes $X_{\mathcal{A}}(h')$.

Proof. A lifecycle certificate covers every attacked history reachable under the updated controller and admissible attack family. Applying Lemma 1 pointwise gives $\pi_{\theta^+}(h') \in K_\Gamma(h')$ for each such history. This is the definition of $\theta^+ \in \Theta_\Gamma(h)$. A failed membership leaves the updated action map uncertified at that history. An empty kernel additionally requires a changed information set or rejection of certified operation. $\square$

Proposition 3 (Linear-policy parameter gate). *Consider a finite abstraction of future attacked histories indexed by $z \in \mathcal{Z}$ and a linear policy $\pi_\theta(z) = \theta^\top \phi(z)$. Suppose each certificate produces an interval kernel $K_\Gamma(z) = [\ell_z, u_z]$. The lifecycle condition over $\mathcal{Z}$ is then equivalent to*

$$\theta^\top \phi(z) \leq u_z, \quad -\theta^\top \phi(z) \leq -\ell_z, \quad \forall z \in \mathcal{Z}. \quad (12)$$

Optional box constraints on θ preserve a convex polytope. Rejecting candidates outside this set, or projecting onto it, is sound for the finite abstraction.

Proof. For fixed z , membership $\pi_\theta(z) \in [\ell_z, u_z]$ is equivalent to the two inequalities above. Enforcing them for every $z \in \mathcal{Z}$ implements the lifecycle gate on the abstraction. Parameter box constraints preserve convexity. A feasible or projected candidate satisfies every certified interval kernel in the abstraction. $\square$

B. Normalized Reward Influence on a Release Gate

We now connect the reward-record attacker to the halfspaces above. Consider a fixed on-policy batch of length T . Let $J \in \mathbb{R}^{T \times d}$ contain the policy-score row vectors, let $r \in \mathbb{R}^T$ be the logged rewards, and let $H_{ij} = \mathbf{1}[j \geq i]$ be the reward-to-go operator. Define the centering matrix

$$C = I - \frac{1}{T} \mathbf{1}\mathbf{1}^\top \quad (13)$$

and let the attacker choose $\delta \in [-\epsilon, \epsilon]^T$. The centered return vector is

$$y(\delta) = CH(r + \delta). \quad (14)$$

The fixed-batch assumption is appropriate for a reward-record-only attacker: changing a reward after collection does not change the states, actions, or score rows already in the batch.

Proposition 4 (Standardized reward-influence set). *Suppose the learner standardizes the centered reward-to-go advantages with population standard deviation whenever $\|y(\delta)\|_2 > 0$.*

Before optimizer clipping, a gradient-ascent proposal of size α is

$$\tilde{\theta}^+(\delta) = \theta + \frac{\alpha}{\sqrt{T}} J^\top \frac{y(\delta)}{\|y(\delta)\|_2}. \quad (15)$$

Consequently, the raw one-batch proposal set is the image of the centered reward-to-go zonotope under normalization and the policy-score map; in general, it is not a parameter-space zonotope.

Proof. The standardized advantage vector is $y/\sqrt{T^{-1}\|y\|_2^2} = \sqrt{T}y/\|y\|_2$. Substitution into the usual mean score-function gradient $T^{-1}J^\top(\sqrt{T}y/\|y\|_2)$ gives (15). Equation (14) is affine in the reward-box variable, but division by its Euclidean norm is not; hence the normalized image is generally non-affine. $\square$

Theorem 2 (Positive support under global gradient clipping). *Let $B = J^\top/\sqrt{T}$, let the learner cap the global gradient norm at $G > 0$, and assume coordinatewise parameter clipping is inactive over the reward box. The post-cap update before parameter clipping is*

$$\theta_G^+(y) = \theta + \frac{\alpha B y}{\max\{\|y\|_2, \|B y\|_2/G\}}. \quad (16)$$

For a release halfspace $a^\top P \theta \leq b$, define $q = \alpha B^\top P^\top a$, $y_0 = CHr$, and $M = CH$. For any fixed $t > 0$, a reward perturbation whose gate-coordinate increment is at least t exists if and only if the following second-order-cone problem is feasible:

$$\begin{aligned} \lambda &\geq 0, & -\epsilon \lambda \mathbf{1} &\leq v \leq \epsilon \lambda \mathbf{1}, \\ \bar{y} &= \lambda y_0 + M v, & q^\top \bar{y} &= 1, \\ t \|\bar{y}\|_2 &\leq 1, & (t/G) \|B \bar{y}\|_2 &\leq 1. \end{aligned} \quad (17)$$

Feasibility is monotone in t , so bisection on $[0, \min\{\|q\|_2, \alpha G \|P^\top a\|_2\}]$ computes the exact positive support

$$\max \left\{ 0, \sup_{\|\delta\|_\infty \leq \epsilon} \frac{q^\top y(\delta)}{\max\{\|y(\delta)\|_2, \|B y(\delta)\|_2/G\}} \right\}. \quad (18)$$

The construction remains valid when the centered-return zonotope contains the origin; if $G = \infty$, the second norm constraint is omitted. If this support does not exceed the current halfspace margin, no admissible reward edit can cross that halfspace in the batch.

Proof. Global norm clipping gives the stated maximum in the denominator. The ratio is positively homogeneous, so optimization over the zonotope $Z = \{y_0 + M\delta : \|\delta\|_\infty \leq \epsilon\}$ is equivalent to optimization over its conic hull. The first line of (17) is the perspective representation of that hull: $v = \lambda\delta$. This is the standard homogenization used in fractional and conic optimization [14], [15].

If a positive ratio reaches t , rescale its conic representative until $q^\top \bar{y} = 1$; its two denominator branches are then exactly the last line of (17). Conversely, any feasible $\bar{y}$ is nonzero because of the unit-numerator equality and recovers a reward-box ray whose ratio reaches t . Thus the origin cannot create the

degenerate feasibility present in a direct zonotope formulation. Each fixed- t problem is a second-order feasibility-cone problem [16]; monotonicity permits quasiconvex bisection [17]. Cauchy–Schwarz supplies both stated upper bounds. The margin result follows by adding the maximum increment to the current gate coordinate. $\square$

The theorem applies to one fixed batch. Later batches change J , r , and the visited states, producing a sequence of state-dependent normalized updates. The result covers global Euclidean gradient clipping. Coordinatewise parameter clipping remains outside the exact support computation. The implementation also skips standardization below a numerical tolerance, so each reported cone witness must remain above that threshold.

C. A Margin-Robust Finite-Abstraction Lift

Proposition 3 is exact on its listed grid. Lifting that result to continuous reachable histories requires coverage and regularity assumptions. Let $z = \psi(h')$ be a finite-dimensional representation on which the updated policy depends. Suppose the fixed-controller certificate can be written as finitely many obligations

$$c_j(x, u, w, x^+) \leq 0, \quad j = 1, \dots, m, \quad (19)$$

where $x^+ = f(x, u, w)$. Invariant, discrete-time barrier, and bounded-horizon reachability checks can be written in this form after augmenting the state with time or certificate memory.

Theorem 3 (Margin-robust finite-abstraction sufficiency). *Fix a candidate snapshot π_θ and a future-history horizon. Let $\{(z_i, x_{iq}, w_{ir})\}$ be a finite set such that, for every reachable attacked history h' , every $x \in X_{\mathcal{A}}(h')$, and every $w \in \mathcal{W}$, some sampled triple satisfies*

$$\|z - z_i\| \leq \delta_z, \quad \|x - x_{iq}\| \leq \delta_x, \quad \|w - w_{ir}\| \leq \delta_w, \quad (20)$$

where $z = \psi(h')$. Assume:

- 1) $\pi_\theta(z) \in \mathcal{U}$ and $\|\pi_\theta(z) - \pi_\theta(z')\| \leq L_\pi \|z - z'\|$;
- 2) the certificate model f has uniform error $\|f(x, u, w) - \hat{f}(x, u, w)\| \leq \varepsilon_f$;
- 3) $\hat{f}$ is separately Lipschitz with constants L_f^x, L_f^u, L_f^w ; and
- 4) each c_j is separately Lipschitz in (x, u, w, x^+) with constants $L_j^x, L_j^u, L_j^w, L_j^+$.

Define

$$\begin{aligned} \eta_j &= (L_j^x + L_j^+ L_f^x) \delta_x \\ &\quad + (L_j^u + L_j^+ L_f^u) L_\pi \delta_z \\ &\quad + (L_j^w + L_j^+ L_f^w) \delta_w + L_j^+ \varepsilon_f. \end{aligned} \quad (21)$$

If every sampled triple satisfies

$$c_j(x_{iq}, \pi_\theta(z_i), w_{ir}, \hat{f}(x_{iq}, \pi_\theta(z_i), w_{ir})) \leq -\eta_j \quad (22)$$

for all j , then $\pi_\theta(\psi(h')) \in K_\Gamma(h')$ for every reachable attacked history h' in the horizon. Hence the snapshot satisfies the continuous future-history lifecycle condition of Theorem 1.

Proof. Fix h' , $x \in X_{\mathcal{A}}(h')$, and $w \in \mathcal{W}$, and select a sampled triple using (20). Write $u = \pi_{\theta}(z)$ and $u_i = \pi_{\theta}(z_i)$. Policy regularity gives $\|u - u_i\| \leq L_{\pi}\delta_z$. The model-error and Lipschitz assumptions give

$$\|f(x, u, w) - \hat{f}(x_{iq}, u_i, w_{ir})\| \leq \varepsilon_f + L_f^x \delta_x + L_f^u L_{\pi} \delta_z + L_f^w \delta_w. \quad (23)$$

Applying the four Lipschitz bounds of c_j and collecting terms yields

$$\begin{aligned} c_j(x, u, w, f(x, u, w)) &\leq c_j(x_{iq}, u_i, w_{ir}, \\ &\quad \hat{f}(x_{iq}, u_i, w_{ir})) + \eta_j \\ &\leq 0. \end{aligned} \quad (24)$$

The argument holds for every plausible state, disturbance, certificate obligation, and reachable attacked history, which is precisely membership in the corresponding kernels over the stated horizon. $\square$

Theorem 3 converts a justified joint cover into a verification result. Establishing that cover remains a separate obligation; a finite rollout sample is insufficient. The coverage audit in Section VIII provides empirical evidence about this premise without constituting a cover proof.

Proposition 5 (Plausible-set and freeze-frontier monotonicity). *Fix a history domain $\mathcal{H}$, dynamics, certificate obligations, and policy class. If two information contracts satisfy $X_1(h) \subseteq X_2(h)$ for every $h \in \mathcal{H}$, let $K_{\Gamma,k}$ be the kernel induced by X_k and let $\Theta_{\Gamma,k} = \{\theta : \pi_{\theta}(h) \in K_{\Gamma,k}(h), \forall h \in \mathcal{H}\}$. Then*

$$K_{\Gamma,2}(h) \subseteq K_{\Gamma,1}(h) \quad \text{and} \quad \Theta_{\Gamma,2} \subseteq \Theta_{\Gamma,1}. \quad (25)$$

Consequently, for a nested ambiguity family $X_{\rho_1}(h) \subseteq X_{\rho_2}(h)$ whenever $\rho_1 \leq \rho_2$, certified update feasibility is non-increasing in ρ : once the certified parameter set is empty, it remains empty at larger ambiguity. Conversely, a trusted anchor that pointwise shrinks the plausible-state sets can only weakly enlarge the kernels and certified parameter set.

Proof. An action certified for every state in the larger set $X_2(h)$ is certified for every state in its subset $X_1(h)$, proving kernel inclusion. Requiring a policy action to lie in the smaller kernel $K_{\Gamma,2}(h)$ for every $h \in \mathcal{H}$ is at least as restrictive as requiring membership in $K_{\Gamma,1}(h)$, proving parameter-set inclusion. The frontier and anchor statements follow by applying these inclusions to nested families. $\square$

The monotonicity comparison holds on a fixed history domain and certificate contract. If an anchor changes which histories are reachable, or if two sampled plausible sets are not nested, neither conclusion follows automatically. The P3 anchor audit therefore constructs nested samples explicitly.

The three results have distinct scopes. The future-history theorem states a necessary certificate condition. Proposition 3 is exact on its finite abstraction. Theorem 3 supplies a sufficient lift under its cover, regularity, and margin assumptions.

D. Instantiating the Kernel

For an invariant certificate $I \subseteq S$, the kernel is

$$K_I(h) = \{u \in \mathcal{U} : \forall x \in X_{\mathcal{A}}(h), \forall w \in \mathcal{W}, f(x, u, w) \in I\}. \quad (26)$$

For a barrier certificate $B(x) \geq 0$ with safe set $C = \{x : B(x) \geq 0\}$, the corresponding kernel is

$$K_B(h) = \{u \in \mathcal{U} : \forall x \in X_{\mathcal{A}}(h), \forall w \in \mathcal{W}, B(f(x, u, w)) \geq 0\}. \quad (27)$$

A reachability certificate starts from $X_{\mathcal{A}}(h)$ rather than a nominal state estimate. A runtime shield must select an action in $K_{\Gamma}(h)$. An empty kernel requires rejection, fail-safe operation, or a trusted state anchor.

E. A Minimal Counterexample

Consider

$$x_{t+1} = \lambda x_t + u_t, \quad |u_t| \leq u_{\max}, \quad S = [-1, 1]. \quad (28)$$

Suppose FDI makes the same history compatible with $x_t = r$ and $x_t = -r$, where $r \in (0, 1]$. The one-step safe action set for state x is

$$\mathcal{U}_{\text{safe}}(x) = [-u_{\max}, u_{\max}] \cap [-1 - \lambda x, 1 - \lambda x]. \quad (29)$$

For $x = r$ and $x = -r$, these sets are individually nonempty but disjoint whenever

$$1 < \lambda r < 1 + u_{\max}. \quad (30)$$

The strict upper bound makes each known state individually controllable. FDI creates indistinguishability, so no single history-dependent action certifies both states.

This example distinguishes fixed filtering from certified learning. A fixed severe-attack filter can reject when the common kernel is empty. A certified online learner must also prevent updates under a certificate with no feasible kernel.

F. Certificate-Gated Update

The resulting update rule first obtains $X_{\mathcal{A}}(h)$ from a severe-attack-aware estimator or safety filter. It then computes $K_{\Gamma}(h)$. An empty kernel rejects the certificate and freezes the update. Otherwise, the mechanism accepts, projects, constrains, or rejects the candidate according to the updated action's kernel membership.

This rule separates fixed-controller safety from the lifecycle condition for a controller that learns from attacked histories. The underlying certificate continues to supply the controller-synthesis and safety obligations.

VI. DEPLOYMENT CONTRACTS AND ESCAPE CONDITIONS

LifecycleGate is a contract-enforcement layer around a severe-attack-aware filter. It consumes the certified kernel produced by an invariant, barrier, reachability, or shield certificate. The layer then accepts, projects, or rejects a proposed learning update.

Update-admission interface. At history h_t , an estimator supplies $X_{\mathcal{A}}(h_t)$, a certificate module supplies $K_{\Gamma}(h_t)$, and

the learner proposes $\theta^+ = U(\theta_t, h_t)$. The prototype checks the current action against K_Γ or, when a future-history abstraction is available, checks or projects the full update against Θ_Γ . For linear policies this is the convex halfspace system in Proposition 3. A grid certifies only its listed points unless the cover, regularity, and model-error premises of Theorem 3 justify its robust margin.

Contract choices. The mechanism can accept, project, freeze, roll back, or request a trusted anchor that shrinks $X_{\mathcal{A}}$. Commit admission certifies a candidate on a declared state set and horizon. It backtracks along the connected segment from the last admitted snapshot to exclude disconnected safe islands. A permanent filter certifies $F \circ \pi_{\theta^+}$ at runtime. Resident predictive authority repeatedly tests recoverability and switches to the baseline when required. Commit admission incurs coverage and model-mismatch risk; resident mechanisms incur intervention, rejection, and online cost.

Fail-closed semantics. An empty plausible-state kernel, an uncertified snapshot, model failure, or solver failure triggers a fail-closed response. The gate freezes or rolls back and reports the event. A trusted anchor can resume learning by changing the justified plausible-state set and carries an explicit infrastructure cost.

Utility is an independent property. The gate enforces certificate membership independently of the task objective. A safe update may be neutral or harmful. Accepted parameter distance measures update availability, whereas paired task outcomes against always-freeze measure learning utility under benign change.

VII. THIRD-PARTY SHIELD-RETIREMENT CASE STUDY

The controlled contract in Section VIII isolates the mechanism. This section examines the same integrity boundary in an independently published lifecycle that retires runtime authority. Carr et al. [9] (AAAI 2023) train a policy-gradient agent behind a Storm belief-support shield in a partially observable gridworld. Their lifecycle later removes the shield either suddenly or smoothly and compares the released policy with retained-shield and no-shield controls. We use the official public implementation and modify only learner-visible reward records. The attacker writes $r' = r + \xi$, with $|\xi| \leq \delta$ per record, before the trusted update. States, observations, shield semantics, dynamics, learner code, hyperparameters, and evaluation follow the upstream implementation. The artifact itemizes every instrumentation and attack patch. REINFORCE training runs for 5000 episodes, with retirement at episode 1000. The TF-Agents proximal policy optimization (PPO) pipeline retires the shield after 10^5 environment steps.

a) Lifecycle fidelity.: On the obstacle domain ($N=6$), the clean condition reproduces the qualitative ordering of Carr et al.’s Table 1 for REINFORCE: shield retained 0/0 violations during/after training, smooth 326/675, sudden 3277/3784, and no-shield 3798/3784. Shield < smooth $\ll$ sudden $\leq$ no-shield is the ordering reported by Carr et al., establishing the lifecycle fidelity of the instrumented implementation.

TABLE II
RETIREMENT-BOUNDARY CAUSAL ISOLATION ON THE THIRD-PARTY LIFECYCLE (OBSTACLE, REINFORCE, SUDDEN). EACH ROW IS A PROVENANCE-COMPLETE PAIRED TRAINING RUN. POISONING STOPS AT RETIREMENT, AND ALL LATER TRAINING IS CLEAN. AT-RETIREMENT EVALUATES THE RAW POLICY AT EPISODE 1000; FINAL EVALUATES IT AT EPISODE 5000. RATES ARE VIOLATION FRACTIONS.

configured seed	at retirement		final	
	clean	poisoned	clean	poisoned
1	0.216	0.777	0.236	0.235
2	0.484	0.763	0.234	0.751
3 ^a	0.241	0.242	0.747	0.228

^aAn earlier run-index-3 clean summary duplicated a separate fix-check record and is excluded. This row reports a fresh clean/poisoned pair generated under the locked configuration. The archived poisoned realization (0.748 at retirement) lacks a provenance-valid clean endpoint and is excluded from paired inference.

b) Causal isolation at the retirement boundary.: Under a locked protocol (artifact amendment v1.3), poisoning is confined to the shielded bootstrap phase. Reward-record mutation is active only while the shield enforces the action mask and stops at retirement. All subsequent learning is clean. We evaluate the raw policy at the switch instant before any unshielded learning (1000 episodes) and again after the full training budget (5000 episodes). Each independently trained clean/poisoned pair is one training-level replicate; configured seeds serve only as run indices. Table II and Fig. 2 report the provenance-complete pairs. A pair is *positive* when the poisoned at-retirement rate exceeds its clean pair by at least 0.15 and the paired McNemar exact test gives $p < 0.01$. All other outcomes are *non-positive*; *protective* denotes a lower poisoned rate.

Two of the three provenance-complete pairs had higher raw-policy violation rates at retirement under poisoning, by factors of 3.6 and 1.6. The fresh third pair was unchanged ($0.241 \rightarrow 0.242$). Physical violations remained zero throughout protected execution. The positive outcomes located latent policy damage before unshielded learning. The fresh pair showed that the fixed attack can fail in a low-disagreement retirement state. After 4000 episodes of clean post-retirement learning, one positive pair recovered (0.235 vs. clean 0.236), one remained worse (0.751 vs. 0.234), and the fresh non-positive pair reversed (0.228 vs. 0.747). We denote clean raw-policy/shield disagreement at retirement by D_{ret} . In the two positive pairs, D_{ret} was 0.025 and 0.047 and rose to 0.113 and 0.109 under poisoning. The fresh non-positive pair also had low clean disagreement ($D_{\text{ret}} = 0.013$), separating available headroom from attack success.

c) Learner boundary.: The same contrast attack ($\delta=2$) was evaluated on the third-party PPO pipeline with action masking and sudden retirement. It raised post-removal violations in one of three paired runs, from 1178 to 2439 (McNemar exact $p = 2.6 \times 10^{-149}$). The first-violation episode changed from 6 to 0. The other two runs were unchanged ($p = 1.00$ and 0.96). In a separate ten-run shield-on-only

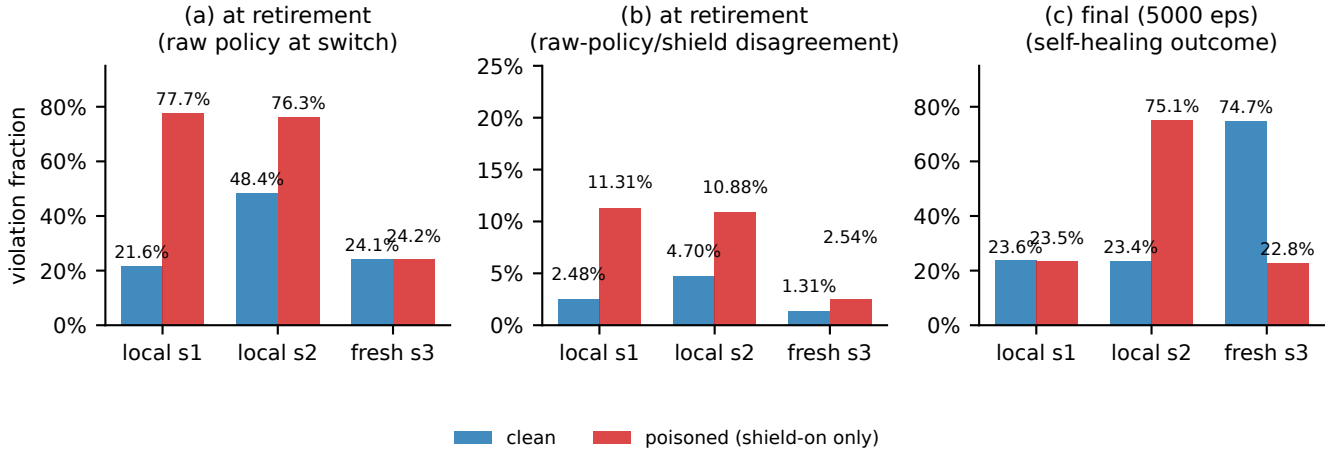


Fig. 2. Causal isolation on the third-party lifecycle (REINFORCE, obstacle, sudden, provenance-complete paired runs). (a) At-retirement violation fraction of the raw policy; (b) at-retirement raw-policy/shield disagreement; (c) final violation fraction after clean post-retirement learning. Poisoning is confined to the shielded bootstrap phase. The run-index-3 bars show the fresh pair in Table II.

isolation battery, the run indexed by configured seed 101 was positive ($0.219 \rightarrow 0.754$ at retirement; McNemar exact $p = 3.7 \times 10^{-113}$), while the other nine entered a high-violation ceiling regime (clean $0.728\text{--}0.760$) and were either protective or unchanged. Figure 3 reports the full battery. A ten-run full-phase PPO battery likewise placed every clean policy near the retirement ceiling ($0.73\text{--}0.77$) and yielded no positive outcome on the primary retirement metric. Its final metric captured a different effect: all ten poisoned policies remained unsafe ($0.49\text{--}0.77$), whereas nine clean policies recovered to approximately 0.22 under continued learning.

A ten-run shield-on-only battery on off-policy soft actor-critic (SAC) provided a complementary boundary. Three runs were positive ($0.207 \rightarrow 0.561$, $0.218 \rightarrow 0.513$, and $0.479 \rightarrow 0.954$; paired McNemar exact $p < 10^{-40}$ for each), three were protective, and four were unchanged. All ten clean SAC policies had low retirement disagreement rather than occupying the ceiling regime. Four extension runs generated on a second host were non-positive. The fixed attack therefore affected individual PPO and SAC training realizations without producing reliable success within either family.

d) Attack budget and state dependence.: On the full-phase REINFORCE protocol, the effect varied with attack shape, budget, and training realization. In the realization indexed by configured seed 2, $\delta=2$ raised post-removal violations from 1127 to 3762 ($3.3\times$; McNemar exact p below numerical reporting precision). Increasing the budget to $\delta=10$ produced 3766 violations. In the realization indexed by seed 3, $\delta=2$ was unchanged (2482 vs. 2471; $p = 0.84$), whereas $\delta=10$ raised violations to 3700 ($p = 8.4 \times 10^{-138}$). The realization indexed by seed 1 began near the unsafe ceiling (75.7%) and showed no added damage at either budget. Shape controls were also state-dependent. The bias variant was protective in the ceiling realization. The risk variant was protective in two realizations and raised violations from 1127 to 2457 in the

realization indexed by seed 2.

An independent Linux-host batch with identical dependency versions contained one low-clean-violation realization and two ceiling realizations. The former showed the contrast effect at retirement ($0.232 \rightarrow 0.501$) and after removal ($1157 \rightarrow 2451$; $p = 4.6 \times 10^{-159}$). The latter two were unchanged at $\delta=2$ and protective at $\delta=10$. The run index of the low-clean-violation regime differed across hosts. Repeating the clean configuration also entered a different trajectory mode (0.776 rather than 0.232 at retirement). Dose comparisons are therefore confined to the locked local batch. The independent batch shows that attackability follows the realized retirement state more closely than the nominal seed identifier.

e) Escape condition.: Under the same full-scope poisoning used in the full-phase battery, a retained shield kept physical violations at zero before and after training in all three paired runs. The corresponding sudden-retirement runs exposed final violation fractions of 0.490, 0.752, and 0.744 after removal. Resident authority therefore contained the downstream consequence of full-scope contamination; removing it exposes the same consequence as an unsafe released policy.

f) Interpretation.: Carr et al.’s workflow supplies the lifecycle component: protected bootstrap followed by retirement of the shield’s action authority. The controlled study in Section VIII supplies the adversarial-provenance mechanism. Their composition exposes an integrity boundary at authority transition. Carr et al.’s original evaluation concerns shield-removal behavior under unpoisoned learning; the case study concerns evidence reuse across the same transition under adversarial update provenance. Absolute rates differ across instrumented and non-instrumented code versions because the unmasked-disagreement probe requires an independent belief tracker; all clean-vs-poisoned comparisons above are paired within a locked code version and are unaffected by that version change.

Training outcomes also varied across platforms. In a ten-

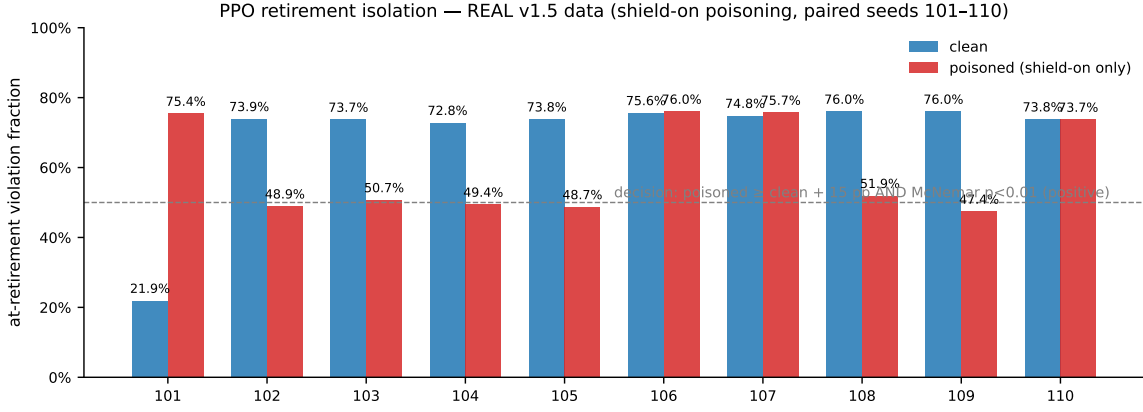


Fig. 3. PPO retirement-boundary isolation on the third-party lifecycle (shield-on-only contrast poisoning, $\delta=2$, paired runs indexed by run indices 101–110, sudden retirement). At-retirement violation fraction of the raw policy for clean and poisoned runs. The run indexed by 101 is positive ($0.219 \rightarrow 0.754$; McNemar exact $p = 3.7 \times 10^{-113}$); the other nine runs land in the unsafe ceiling regime (clean at-retirement $0.728\text{--}0.760$) with no paired headroom, six protective by the paired test and three unchanged. The dashed line marks the locked decision boundary (poisoned at least 0.15 above clean with paired McNemar $p < 0.01$).

Budget- and state-dependent susceptibility (full-phase REINFORCE, paired within locked code version vA)

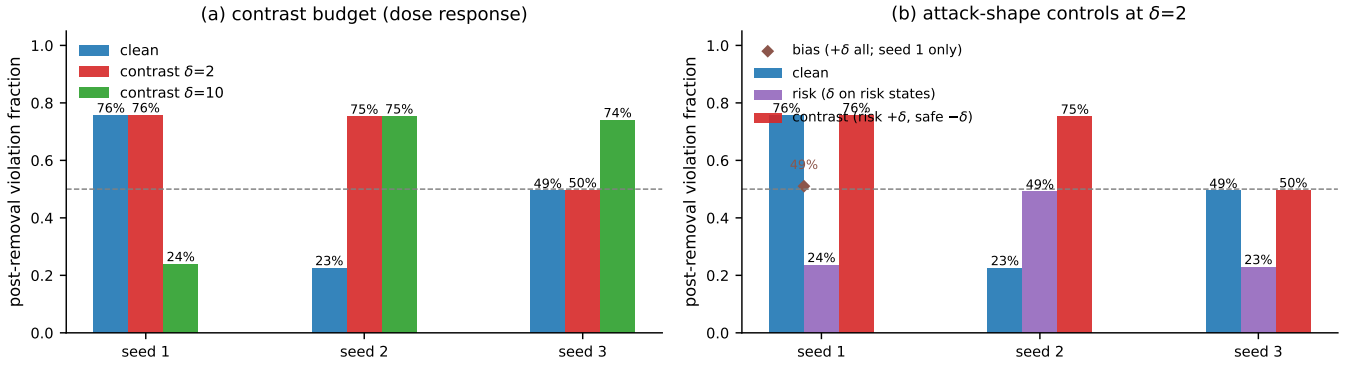


Fig. 4. Budget- and state-dependent effects on the full-phase REINFORCE protocol (paired within one locked code version). (a) Contrast attack dose response; (b) attack-shape negative controls at $\delta=2$. The run-index-2 realization saturates at $\delta=2$; the run-index-3 realization requires $\delta=10$; the ceiling realization shows no additional damage at either budget, and bias/risk controls are protective or strictly weaker.

run REINFORCE shield-on-only battery on Linux, three low-clean-violation realizations were positive ($0.240 \rightarrow 0.533$, $0.234 \rightarrow 0.744$, and $0.229 \rightarrow 0.763$), while seven ceiling realizations showed no added damage. By contrast, the fresh pair in Table II was non-positive despite low clean disagreement. The same configured seed and code can enter different trajectory modes across hosts or backends. We therefore treat each independently trained clean/poisoned pair as a replicate and use nominal seeds only as run indices. Episode-level McNemar tests describe outcomes conditional on a trained pair; the multi-run batteries provide the training-level replication evidence.

The broader cross-tabulation reinforced this state dependence. Across 53 locked paired runs spanning both scopes, all three learner families, and both domains, 12 of 20 runs with

clean at-retirement violation below 0.5 were positive, whereas none of 33 ceiling runs was positive (Fig. 6). The eight low-rate exceptions comprise six SAC runs and two protective runs in the second environment. Low clean violation therefore marks available headroom but does not determine the attack outcome.

The 50 obstacle runs with disagreement instrumentation provided a sharper retirement-state marker. Clean D_{ret} was strongly associated with clean violation (Spearman 0.976 , $p = 1.3 \times 10^{-33}$) and negatively associated with attack headroom (Spearman -0.547 , $p = 3.9 \times 10^{-5}$). Every positive outcome had clean $D_{\text{ret}} \leq 0.047$, while non-positive outcomes extended to 0.111 . Under the fixed bands, all 11 positive outcomes fell in the low-disagreement region ($D_{\text{ret}} < 0.06$), and none of 32 high-disagreement runs was positive. The

area under the receiver operating characteristic curve (AUC) was 0.953 (Mann–Whitney $p = 5.5 \times 10^{-6}$). Family-stratified counts clarify this statistic. REINFORCE had seven positive low-band runs and 13 non-positive high-band runs. PPO had one positive low-band run and 19 non-positive high-band runs. All ten SAC runs were low-band, with three positive and seven non-positive outcomes. Thus high disagreement acted as a strong exclusion condition in this corpus, whereas low disagreement defined an opportunity region. Eleven of 18 low-band corpus runs were positive; the fresh pair, analyzed separately from the 50 disjoint namespaces, gave an additional low-band non-positive realization.

The protocol chronology bounds the statistical interpretation. The 0.06/0.09 bands were fixed after analysis of 46 existing obstacle rows and before four SAC extension pairs with run indices 407–410. All four extensions were low-band and non-positive. They strengthened the non-sufficiency result without prospectively testing high-disagreement exclusion. Accordingly, AUC 0.953 describes discrimination within the combined locked corpus, not external classifier validation. Because D_{ret} is measured after the shielded bootstrap, it characterizes the realized retirement state. An online attack trigger would require earlier disagreement trajectories that predict the eventual opportunity region. The marker is also obstacle-scoped. Two protective runs in the second environment had clean disagreement 0.476 and 0.157 despite clean violation below 0.5.

g) *Environment boundary.*: On a second gridworld of the same third-party suite (surveillance, $N=6$, $R=3$), a clean reproduction again satisfied the authors’ lifecycle ordering: shield retained 0/0, smooth 0.129, sudden 0.246, and no-shield 0.252 (retained $<$ smooth $\ll$ sudden $\leq$ no-shield). The fixed contrast attack at $\delta=2$ raised post-removal violations in one of three independent paired runs ($0.000 \rightarrow 0.272$, McNemar exact $p = 2.6 \times 10^{-82}$) and was protective in the other two ($0.267 \rightarrow 0.000$, $p = 8.4 \times 10^{-81}$; $0.135 \rightarrow 0.038$, $p = 1.3 \times 10^{-15}$). The lifecycle ordering extended to the second environment, whereas the exploit effect remained environment- and realization-dependent.

VIII. CONTROLLED MECHANISM STUDY

The third-party case study located the authority-transition vulnerability in an independently published lifecycle. This section isolates its mechanism in Safe-Control-Gym [18]. The controlled experiments test whether finite release evidence supports both permanent raw release and resident predictive authority. They also measure reward influence, cross-system failure, integrity controls, and the costs of alternative deployment contracts.

The evidence proceeds in three stages. Low-dimensional semantic benchmarks first distinguish current-action filtering from update admission. A prospective contract-separation audit then compares one-time raw release with repeated recoverability checks and a one-step resident negative control. Finally, cross-system, detectability, and runtime-cost studies

TABLE III
CERTIFIED KERNEL FOR THE ONE-DIMENSIONAL AMBIGUITY SET.

ρ	$\lambda\rho$	$K(\rho)$	Gate
0.600	0.720	$[-0.200, 0.200]$	allow
0.800	0.960	$[-0.040, 0.040]$	constrain
0.833	1.000	$\{0\}$	boundary
0.900	1.080	$\emptyset$	freeze/reject
1.000	1.200	$\emptyset$	freeze/reject

characterize the scope and operational consequences of the fixed attack.

A. Setup

We instantiated the system in Section V with $\lambda = 1.2$, $u_{\max} = 0.2$, and $S = [-1, 1]$. FDI induced the ambiguity set $X_{\mathcal{A}}(h) = [-\rho, \rho]$. The common certified kernel for invariant certificate $I = S$ is

$$K(\rho) = [-u_{\max}, u_{\max}] \cap [-1 + \lambda\rho, 1 - \lambda\rho]. \quad (31)$$

The kernel is nonempty exactly when $\lambda\rho \leq 1$.

B. Learning-Freeze Frontier

The frontier occurs at $\rho = 1/\lambda = 0.833$. Table III shows the exact kernel values for representative ambiguity radii.

At $\rho = 0.9$, states $+0.9$ and -0.9 are individually controllable when their sign is trusted. Their safe intervals are $[-0.2, -0.08]$ and $[0.08, 0.2]$, respectively. The empty intersection prevents a common history-dependent action from being certified.

C. Trusted Anchors and Learning-Aware FDI

A trusted state anchor can restore learning by shrinking the plausible-state set. At $\rho = 0.9$, a trusted sign anchor reduces the set to $[0, 0.9]$ or $[-0.9, 0]$. The corresponding kernels, $[-0.2, -0.08]$ and $[0.08, 0.2]$, are nonempty.

Learning-aware FDI creates the complementary failure mode. For a candidate action $u = 0.1$, the gate allows the update at $\rho = 0.6$, projects it to 0.04 at $\rho = 0.8$, and rejects it at $\rho = 0.9$. The attacked history therefore determines whether the same candidate is allowed, constrained, or rejected.

D. Lifecycle Benchmark Results

We compared five mechanisms over five update steps: an ungated learner, a nominal action filter, a robust action filter, always-freeze, and LifecycleGate with projection. The one-dimensional benchmark used the system above. The linear tank benchmark used the level-deviation model

$$x_{t+1} = Ax_t + Bu_t, \quad A = \begin{bmatrix} 1.08 & 0.04 \\ 0.02 & 1.03 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, \quad (32)$$

with $|u_t| \leq 0.15$ and $x_t \in [-1, 1]^2$. The nonlinear tank benchmark uses square-root inter-tank and drain flows,

$$\begin{aligned} h_{1,t+1} &= h_{1,t} + \Delta t(q_0 + u_t - q_{12}(h_t) - d_1(h_{1,t})), \\ h_{2,t+1} &= h_{2,t} + \Delta t(q_{12}(h_t) - d_2(h_{2,t})), \end{aligned} \quad (33)$$

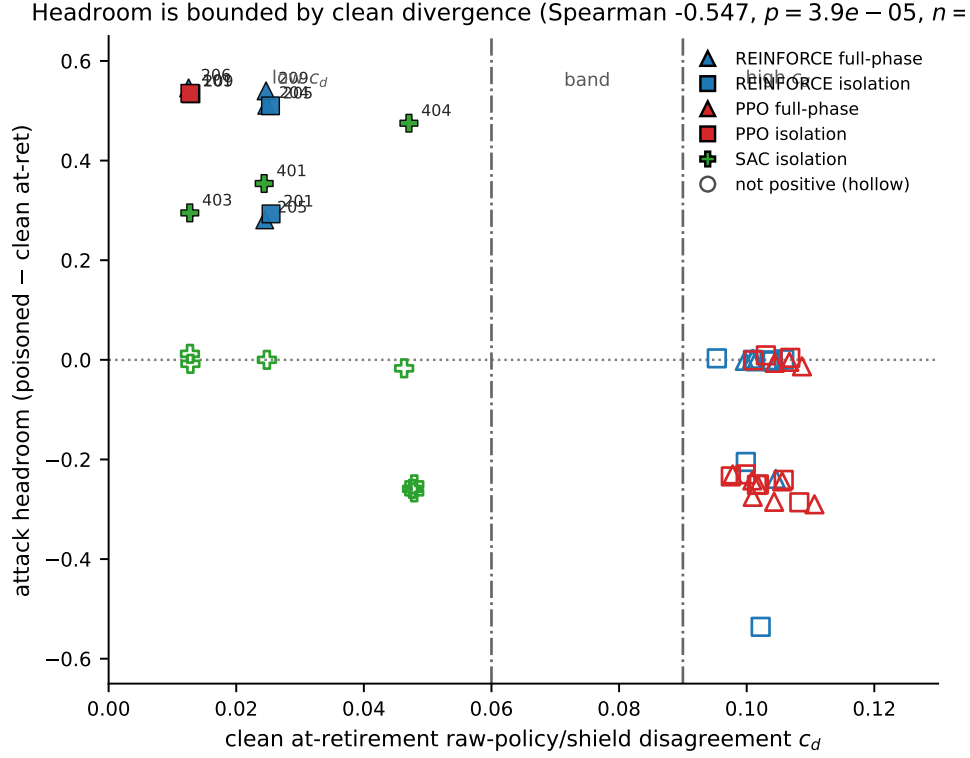


Fig. 5. Clean at-retirement raw-policy/shield disagreement marks an opportunity region across the obstacle corpus (both scopes, all three learner families). Each point is one paired clean/poisoned run; filled markers meet the positive-outcome rule, and hollow markers do not. Filled markers all have clean disagreement ≤ 0.047 ; the dashed lines mark the fixed low/high disagreement bands (< 0.06 and ≥ 0.09). No high-disagreement run is positive, while seven of 18 low-disagreement runs are non-positive.

where q_{12} and d_i are square-root flow terms. Its interval observer returned a box around the nominal attacked history. The one-step kernel was computed over a grid within that box. In all benchmarks, learning-aware FDI pushed the learner toward a larger positive action parameter.

Figure 7 separates current-action filtering from update certification. At $\rho = 0.8$, the nominal filter had unsafe certified rate 1.0. The robust action filter had stale-policy rate 1.0 because the learner parameter left $K(\rho)$ after action projection. LifecycleGate had zero unsafe or stale operation and retained one of five updates. At $\rho = 0.9$, the kernel was empty and LifecycleGate froze learning.

The linear tank benchmark reproduced the same pattern. At $\rho = 0.8$, nominal filtering had unsafe certified rate 0.8, and robust action filtering had stale-policy rate 0.8. LifecycleGate kept both rates at zero and allowed two of five updates. It allowed one constrained update at $\rho = 0.85$ and froze at the empty-kernel point $\rho = 0.92$.

The nonlinear tank benchmark preserved the separation with kernels computed from sampled nonlinear dynamics. At $\rho = 0.2$, nominal filtering was unsafe in four of five steps, and robust filtering was stale in four. LifecycleGate kept both rates at zero while allowing two constrained updates. At $\rho = 0.45$, the interval-observer box admitted no common safe action and

LifecycleGate froze the update.

E. Parameter-Level Update Gate

We also instantiated the gate at the parameter level for linear policy $u = kz + b$. Each future attacked observation z induced a safe-action kernel over ambiguity interval $[z - \rho, z + \rho]$. Requiring $kz + b$ to lie in every kernel produced linear halfspace constraints on (k, b) . The parameter gate projected poisoned candidates onto this certified set.

Figure 8 shows the parameter-level frontier. At $\rho = 0.7$, current-action filtering had stale-policy rate 1.0. The parameter gate kept the rate at zero and permitted two of five updates. It permitted one constrained update at $\rho = 0.8$ and froze at $\rho = 0.85$.

We applied the same construction to policy $u = k_1 z_1 + k_2 z_2 + b$. Each grid observation induced a plausible-state box and a one-step safe-action interval. These intervals produced halfspaces in (k_1, k_2, b) . At $\rho = 0.5$, current-action filtering had stale-policy rate 1.0. The parameter gate kept the rate at zero and admitted all five projected updates.

F. Adaptive Attacker Stress Test

We next allow a finite-search white-box attacker to select among poisoned update directions. Its objectives maximize stale-policy, unsafe-action, or forced-freeze outcomes. For

Susceptibility rule: positive only when clean at-ret < 0.5

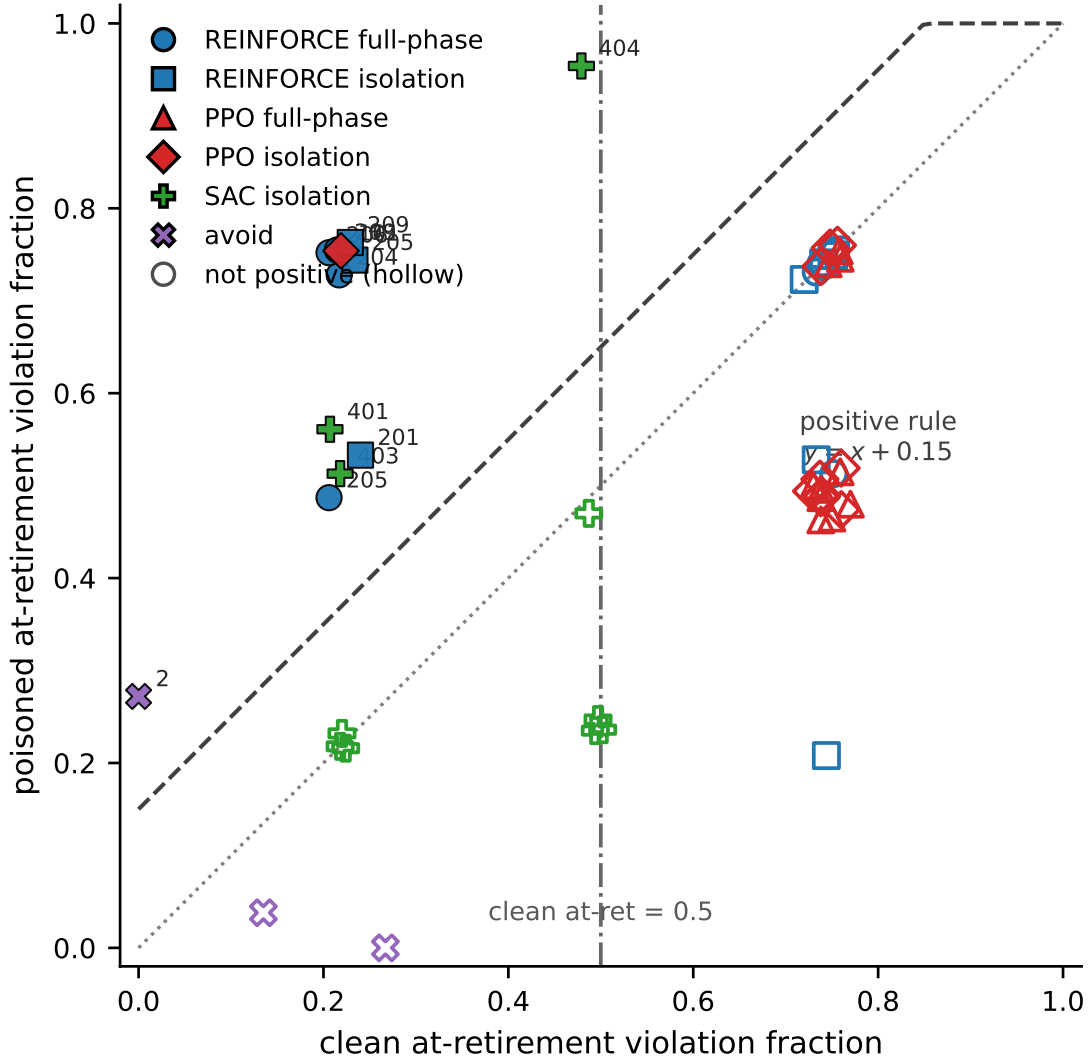


Fig. 6. Two-by-two opportunity-region cross-tabulation across all locked paired runs (both scopes, all three learner families, both domains). Each point is one paired clean/poisoned run; filled markers meet the positive rule (poisoned at-retirement at least 0.15 above clean at retirement with paired McNemar exact $p < 0.01$), hollow markers do not. The dashed rule line is $y = x + 0.15$ and the vertical dashed line marks the clean at-retirement 0.5 boundary: positive outcomes concentrate below 0.5, and no ceiling run is positive.

forced freeze, the attacker also selects the budget-feasible ambiguity radius that most reduces the certified set.

Figure 10 preserves the lifecycle separation under adaptive direction search. At high ambiguity, the robust action filter protected the applied action but left the raw policy stale. LifecycleGate kept unsafe, stale, and uncertified-learning rates at zero while allowing feasible projected updates. At $\rho = 0.84$, the forced-freeze objective made the common kernel empty. The robust filter recorded five uncertified learner updates, whereas LifecycleGate performed none.

G. Safe-Control-Gym Cartpole Semantics

We next instantiate the lifecycle condition in the Safe-Control-Gym cartpole benchmark. The official linear-quadratic regulator (LQR) is augmented with a two-parameter linear residual learner. The attacker injects continuous observation FDI and optimizes a cross-entropy-method (CEM) objective over stale-policy, unsafe-action, and forced-freeze margins. The attacked observation z and radius ρ induce the plausible box $X_A(z) = \{x : \|x - z\|_\infty \leq \rho\}$. We sample this box and evaluate the official CasADi cartpole dynamics over an

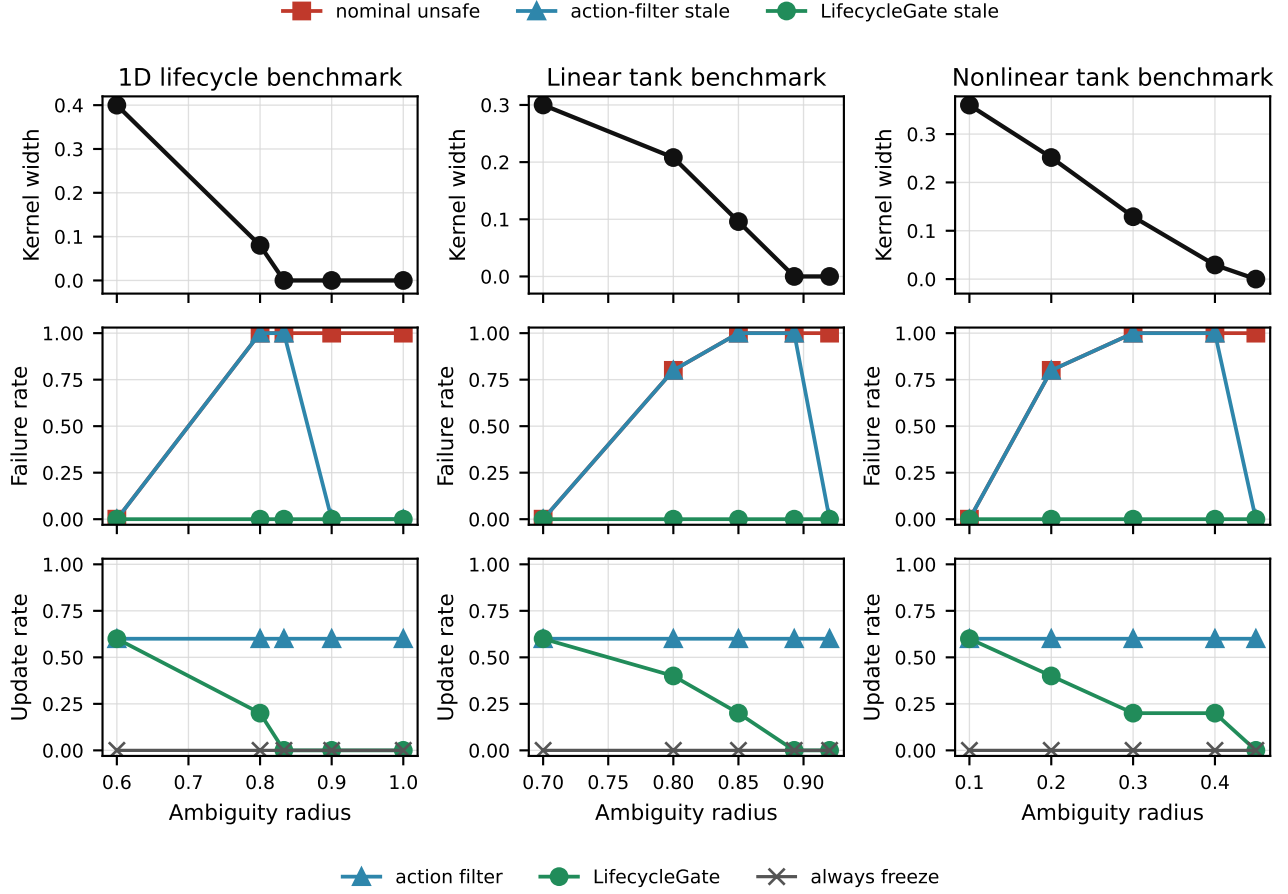


Fig. 7. Lifecycle-gate benchmark results. Increasing ambiguity shrinks the certified kernels of the one-dimensional, linear-tank, and nonlinear-tank models. LifecycleGate constrains or freezes updates when action-only filters would leave the updated policy outside the kernel.

action grid to compute $K_{\Gamma}(z)$. A reference run uses the official cartpole CBF and its sampled Lie-derivative condition.

This experiment is a direct semantic implementation of Theorem 1 and Proposition 3. The future-history abstraction $\mathcal{Z}$ is a local grid of future attacked observations around the current attacked observation. For each $z' \in \mathcal{Z}$, the robust kernel produces an interval $K_{\Gamma}(z') = [\ell_{z'}, u_{z'}]$. The LQR term is fixed, while the residual learner is linear in $(\theta_{\text{gain}}, \theta_{\text{bias}})$, so requiring the updated residual policy to stay inside all future kernels yields the halfspace constraints used by LifecycleGate.

Figure 11 separates three effects. Action filtering removed current constraint violations without certifying the updated residual policy. Always-freeze avoided stale certification but admitted no learning. LifecycleGate projected feasible poisoned updates into the future-history halfspace set, preserving update availability before the freeze frontier. At larger ambiguity radii, the kernel became empty and LifecycleGate froze the update. These results instantiate the certificate semantics in a standard control benchmark; the evaluation does not represent an industrial deployment.

H. Prospective Contract-Separation Audit

All results in this subsection assume the reward-ingestion interface in Table I. Development grids selected target $(18, -5)$ using five-step acceptance and 120-step harm. Before the sequential V3 audit, we fixed the target, learner, state pools, stop criterion, and bounded perturbation $2 \tanh(\epsilon_t(\mu_t^* - \mu_t))$. The run indexed by seed 2070 was used for development. The runs indexed by 2071 and 2072 were then evaluated under the unchanged protocol.

All 72 poisoned snapshot-state pairs pass the initial five-step check. Poisoned permanent raw release then failed on 11, 5, and 7 pairs across the three training runs, versus paired clean counts 0, 0, and 2. Thus 21 of 23 poisoned failures are poison-only, two are shared, and none are clean-only (two-sided exact paired $p = 9.537 \times 10^{-7}$). The CasADi audit identifies the same 23 failures. Resident predictive authority incurs 0/72 failures and switches four to five steps before each corresponding poisoned raw-release failure.

The frozen protocol was then evaluated once on five new learner/evaluation pairs. The target, attacker, learner, budget, state selection, and contracts remained fixed. All 120 snapshot-state pairs passed the initial five-step check. Poi-

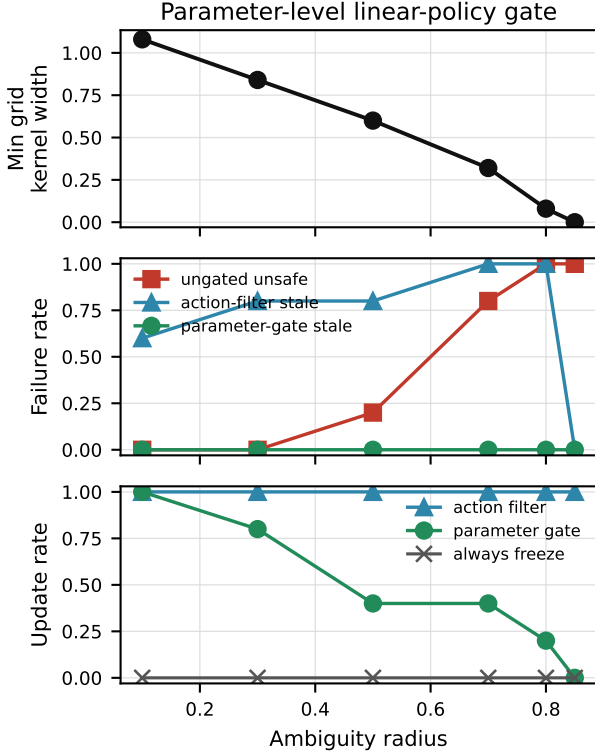


Fig. 8. Parameter-level gate for linear policy $u = kz + b$. Current-action filtering leaves the future action map stale, whereas projection onto the certified halfspace set constrains the full update.

soned raw release failed on 5, 4, 7, 2, and 9 pairs across the five training runs; the paired clean counts were 2, 0, 0, 0, and 0. Resident predictive authority incurred 0/120 failures and switched before all 27 poisoned failures. The 25 poison-only and zero clean-only discordances gave a two-sided exact $p = 5.960 \times 10^{-8}$. Poisoned failure occurred in each of the five independently trained runs.

State-snapshot pairs that share a trained learner are not independent training replicates. The V3 and V4 exact tests therefore summarize within-snapshot discordance. Replication across training realizations is reported separately through the three V3 and five V4 runs.

For independent replication and the one-step negative control, snapshots from run indices 2040–2042 used disjoint evaluation pools 8040–8042. Seventy-one of 72 states passed the initial check, and all mechanisms shared state-snapshot keys.

Table IV makes the three evidence blocks and deployment contracts explicit. A locked horizon sweep reuses the V3 snapshots and states without retraining. Horizons 3 and 5 each retain 72 accepted pairs with 23 poisoned raw failures and zero resident failures; horizon 10 retains 57 accepted pairs, eight poisoned failures across two training runs, and zero resident failures. Thus three horizons met the predeclared separation

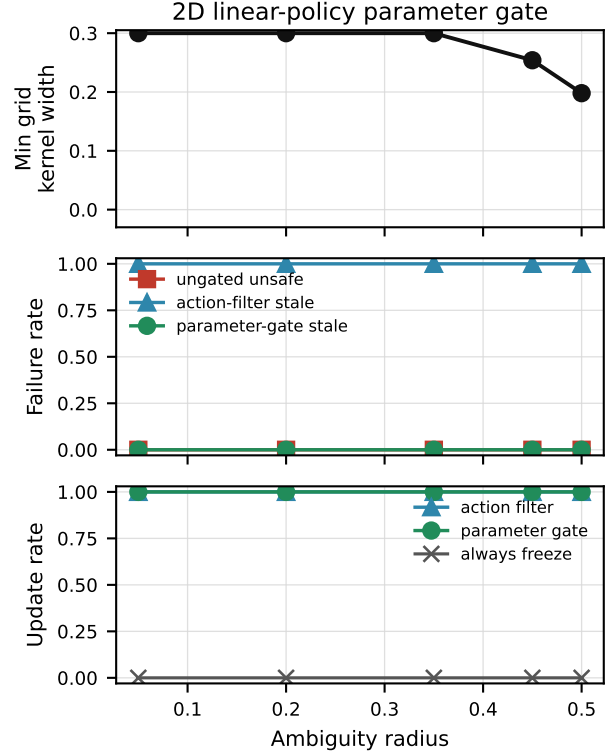


Fig. 9. Parameter-level gate for the two-dimensional linear tank policy. The certified set intersects future-history halfspaces, and the gate projects the poisoned update into that set.

condition. Horizon 1 left 12 resident failures, indicating insufficient lookahead. Horizon 20 admitted 49 pairs but no poisoned failures, indicating conservative rejection.

In the independent audit, permanent raw release failed on 6/24, 3/23, and 10/24 accepted pairs. Resident predictive authority repeats the same five-step raw-policy prediction at every physical step and switches on exactly those 19 paired states, four steps before failure; no physical failure remained. In contrast, the one-step resident kernel failed on 14 pairs despite 79 interventions and reached an empty kernel 41 times. Repeated five-step prediction therefore provided an effective recoverability trigger in this study. One-time authorization and one-step filtering did not provide the same protection.

I. Reward-Influence Geometry and Optimizer Boundary

We first validated the reward-influence analysis on an unclipped cartpole batch from the learner indexed by seed 2040. Reconstructing the implemented gradient from Equation (15) has error 2.78×10^{-17} . The clean gradient norm is 0.196789, neither gradient nor parameter clipping is active, and the minimum centered-return norm over the reward box is 7.993864. Across 22 release-gate halfspaces, the largest discrepancy between an SOCP witness and its computed support is 1.07×10^{-9} . The minimum robust margin was 4.598004, so no halfspace could be crossed in this batch. The

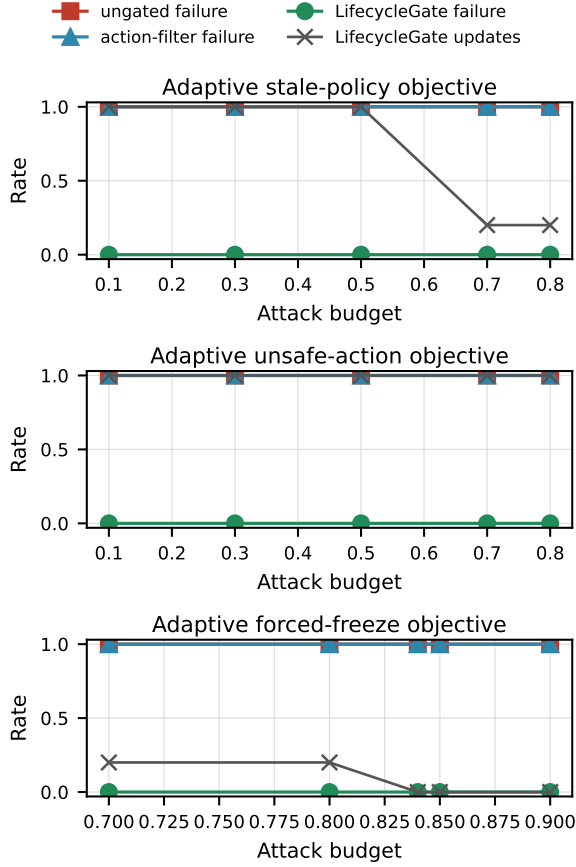


Fig. 10. Finite-search attacker against the scalar parameter gate. The attacker selects update directions or ambiguity radii for three objectives. LifecycleGate freezes when the certified kernel is empty.

successful physical attack instead used the locked bounded tanh target-shaping rule and accumulated influence across changing batches. It was not generated by a one-step second-order cone program (SOCP).

The homogenized clipped formulation is then audited on all 36 poisoned V3 batches. The old premises make 0/36 eligible; the new formulation yields verified support witnesses on 22/36, including 18 whose reward boxes contain a normalization singularity and 20 for which the old global gradient-clipping exclusion fails. Every actual update is below its computed support, all witnesses respect the reward budget, final parameter reconstruction error is zero, and maximum witness/support error is 3.67×10^{-5} . Ten batches cannot globally exclude coordinatewise parameter clipping (nine activate it on the observed edit), and four cone-optimal directions have no recovered witness above the learner’s normalization tolerance; they remain outside the exact implementation audit. The clipped quadrotor trajectories have not received this separate parameter-clipping audit and remain empirical.

Exact batch support did not explain the successful multi-batch attack. The locked tanh update advances in the target direction on only 6/20 eligible batches with positive oracle

TABLE IV
CONTRACT SEPARATION AFTER INITIAL FIVE-STEP ACCEPTANCE UNDER BOUNDED REWARD-RECORD WRITE ACCESS. V4 IS THE FIVE-RUN CONFIRMATION; V3 CONTAINS THE DEVELOPMENT AND SEQUENTIAL AUDITS. THE INDEPENDENT AUDIT INCLUDES THE ONE-STEP NEGATIVE CONTROL. A SWITCH TRANSFERS CONTROL TO THE TRUSTED LQR.

Contract	Failures	Authority event
<i>V4 confirmation (120 pairs; 5 training runs)</i>		
Clean permanent raw release	2/120	—
Poisoned permanent raw release	27/120	—
Resident predictive authority	0/120	27 switches
<i>Development/sequential V3 (72 paired states)</i>		
Clean permanent raw release	2/72	—
Poisoned permanent raw release	23/72	—
Resident predictive authority	0/72	23 switches
<i>Independent audit (71 accepted states)</i>		
Permanent raw release	19/71	—
One-step resident kernel	14/71	79 int.; 41 empty
Resident predictive authority	0/71	19 switches

increment, and its preregistered same-trajectory advantage reaches 22/36 rather than the required 27/36. The original exact-support audit emitted 0/12 edits under the earlier singularity criterion. A diagnostic rerun on the development realization used the revised solver and emitted nonzero edits in 9/12 batches. Its accepted raw snapshot still caused 0/24 release failures. The exact-support result therefore serves as a batch audit. The physical experiment evaluates the fixed target and bounded attacker without establishing joint gate optimality of the tanh rule.

J. Cross-System Reward-Record Poisoning

We next tested the lifecycle failure with updates produced by an online learner. The policy was a Gaussian residual actor on top of a trusted controller. Standard reward-to-go REINFORCE gradients were computed from PyBullet transitions. The attacker changes only the scalar reward written to the update log, subject to a per-step L_∞ budget; it cannot write the action, gradient, parameter, optimizer state, dynamics, or deployment state. A backup action filter protects the adaptation trajectory. Deployment then audits the raw learned snapshot, the object requiring lifecycle admission.

Training, certificate, calibration, and deployment states used disjoint run indices. All mechanisms within a training realization shared paired state sets. Attack directions and hyperparameters were fixed on a development realization before the three-run audits. The exact tests operate at the paired deployment-state-block level, conditional on the three learned snapshots. Their p -values summarize within-snapshot discordance. Wilson and paired-block bootstrap intervals have the same conditional scope. Replication across learners is reported through the independent training realizations.

On cartpole, the clean learner violates 1/188 admitted rollouts and the poisoned action-only learner violates 38/188. There are 37 paired poison-only failures and no clean-only failures (two-sided exact paired $p = 1.455 \times 10^{-11}$). Both

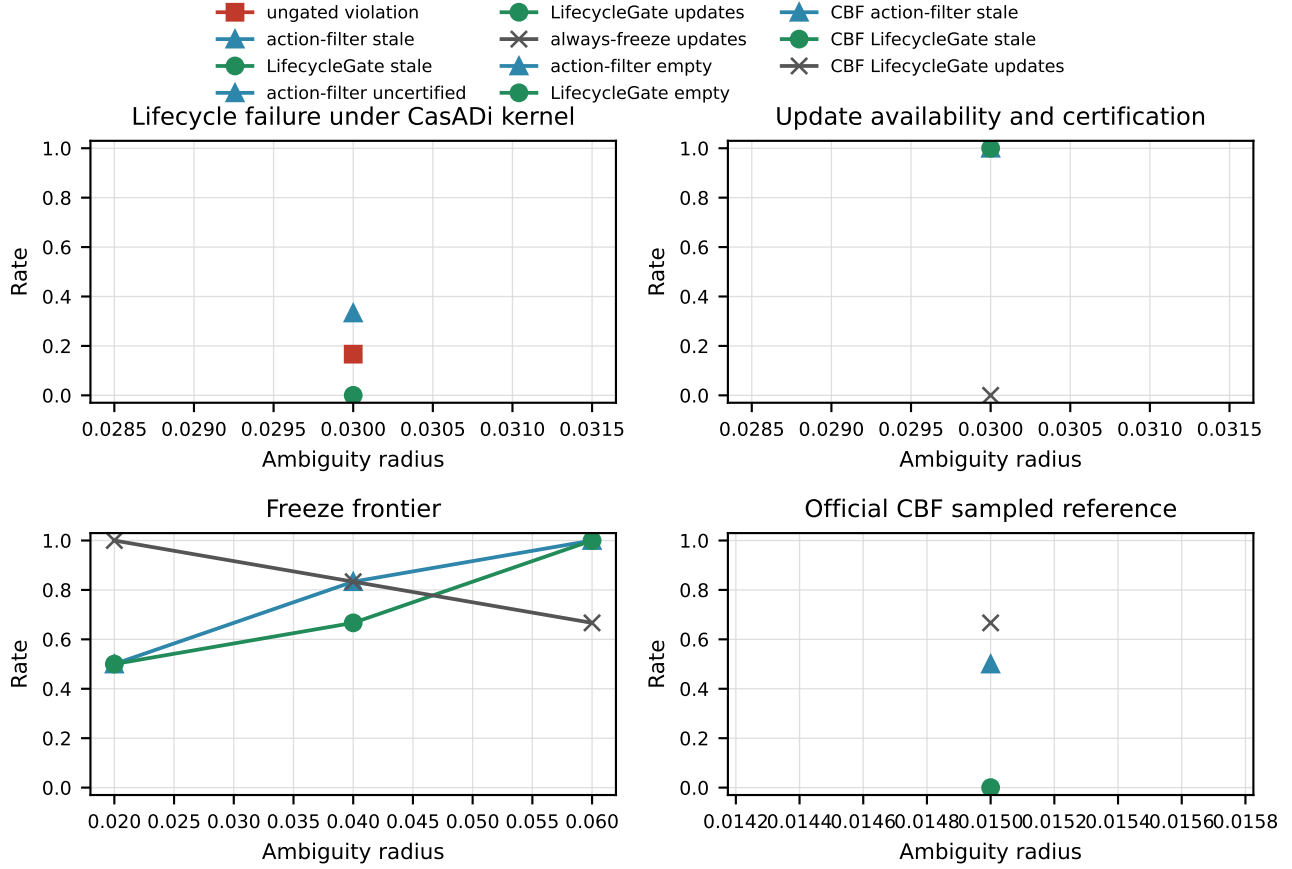


Fig. 11. Safe-Control-Gym cartpole lifecycle evidence from a two-run sweep. The panels compare the CasADi robust kernel, an Euler attack-search surrogate, and a sampled CBF reference. At $\rho = 0.03$, LifecycleGate eliminated stale and uncertified updates while retaining update availability. The kernel became empty as ambiguity increased.

commit admission and always-freeze have zero observed violations.

On two-dimensional quadrotor, clean learning has 0/72 violations, while reward-only poisoning causes 68/72, all delayed, across all three training realizations. The paired discordance is 68 poison-only to zero clean-only (two-sided exact $p = 6.776 \times 10^{-21}$). Commit admission and always-freeze again have zero observed violations. The gate retains nonzero fractions 0.55, 0.40, and 0.70 of the three poisoned pending updates. These fractions establish update availability under attack, not benign learning utility.

Learned-baseline variant. We replaced REINFORCE’s constant batch-mean baseline with a learned per-state linear value baseline under a protocol fixed before execution. The variant retained the same Monte Carlo return, residual actor, score function, and reward-record channel. The trusted learner fitted the baseline from logged rewards. Across three independent training runs, poisoning caused 40/126 raw-release failures, compared with 2/126 for paired clean release. Commit admission and always-freeze remained at 0/126. The 38 poison-only and zero clean-only discordances gave a two-sided exact $p = 7.3 \times 10^{-12}$.

This experiment is a robustness check within the Monte Carlo policy-gradient family. Off-policy and bootstrapped learners change the attack channel through a temporal-difference target. The official-PPO audit below therefore remains the distinct learner boundary. These rates use the full 42-state envelope ($126 = 3 \times 42$) without initial-acceptance filtering and are not directly comparable to the filtered 120-pair confirmation cohort.

Non-singleton adaptation set. The general history model also permits observation FDI (Section III). We therefore added a pole-angle bias with radius ρ to the reward-record attacker. The protective kernel ranged over a plausible box of half-width ρ , approximately $\pm 2.3^\circ$ at $\rho = 0.04$. The release certificate and deployment states remained unchanged. Across three independent training runs, the contract separation persisted at every non-singleton radius. Poisoned raw release failed on 44/126 and 61/126 pairs at $\rho = 0.02$ and 0.04, compared with 4/126 and 3/126 for clean release. Commit admission and always-freeze remained at 0/126. The near-singleton reference at $\rho = 0.005$ was 41/126 versus 5/126. Failure counts were non-monotone across training realizations, while the kernel

TABLE V

PAIRED REWARD-RECORD POISONING AND SNAPSHOT ADMISSION. “COMMIT” DENOTES FINITE-HORIZON ADMISSION; PERMANENT MECHANISMS USE THE COMMON-STATE AUDIT IN FIG. 12. RAW-SNAPSHOT REWARDS AFTER FAILURE ARE DESCRIPTIVE, NOT UTILITY COMPARISONS.

System	Mechanism	Learner seeds	Rollouts	Violations	Mean reward
Cartpole	Clean REINFORCE	3	188	1	−0.197
Cartpole	Poisoned action-only	3	188	38	−1.269
Cartpole	Poisoned commit	3	188	0	−0.102
Cartpole	Always-freeze	3	188	0	−0.117
Quadrotor	Clean REINFORCE	3	72	0	−0.030
Quadrotor	Poisoned action-only	3	72	68	−0.170
Quadrotor	Poisoned commit	3	72	0	−0.060
Quadrotor	Always-freeze	3	72	0	−0.029

TABLE VI

FROZEN REWARD-INTEGRITY AUDIT. RATES ARE BATCH-LEVEL TRUE- AND FALSE-POSITIVE RATES (TPR/FPR) ACROSS SEVEN TRAINING RUNS. EACH CHECK REQUIRES THE STATED DEFENDER ASSUMPTION.

Check	Required trusted information	TPR/FPR
Reward recomputation	Authenticated transition and deterministic reward	1.000/0
Known-sign check	Task guarantee that reward is nonpositive	0.905/0
Scalar envelope	Clean development range transfers	0.917/0.036
Batch-mean envelope	Clean batch mean transfers	0.012/0.036

remained nonempty throughout the grid.

K. Reward-Record Integrity and Detectability Boundary

We evaluated attack detectability separately from deployment failure. Four checks were fixed on the clean development run indexed by seed 2070. The frozen checks were then evaluated on 84 clean and 84 poisoned batches from seven independent training runs.

Table VI shows that trusted recomputation blocks every evaluated edit, and even task-specific scalar checks detect over 90% of poisoned batches. The ineffective batch-mean envelope shows why a generic “monitor the average” statement is not enough: detection depends on reward semantics and trusted inputs. Cryptographic origin binding would also reject post-producer mutation if the trusted producer signs the reward together with its transition, action, and time. That producer remains within the trusted computing base of the evaluated threat model.

We also tested temporal and moment-preserving edits on the development realization. Sparse permutation and projected moment-preserving variants satisfied their integrity constraints and moved parameters toward the harmful target. Each nevertheless caused 0/24 raw-release failures. The effective fixed attack was therefore readily detectable by the task-aware checks. Authentication and trusted reward recomputation provide upstream controls; commit admission and resident predictive authority provide downstream containment.

In-loop known-sign gate. We placed the same known-sign check inside the update loop and froze batches that violated

the nonpositive reward guarantee. Across three independent training runs, this gate reduced poisoned raw-release failures from 39/126 to 0/126 and froze 35 of 36 poisoned batches. The paired defended-versus-undefended test gave $p = 3.6 \times 10^{-12}$. The gate froze no clean batch but approached always-freeze availability under attack. It detects the sign-violating effective attack, while the sign-respecting variants evade the check and independently caused no deployment harm. This gate therefore implements the measured detectability boundary; provenance and recomputation provide the broader upstream controls.

L. Runtime Escape Conditions and Their Cost

Permanent runtime filtering provides a second architectural response. Its certificate covers the composed policy $F \circ \pi$, allowing the filter to contain raw-policy failure before physical execution. We compared permanent and snapshot contracts on a result-independent pool of 144 states from three disjoint run indices. For each training realization, the common operating set contained states certified for both freeze and the committed snapshot at horizon 100 and guard 0.003. Eight order-spanning states were evaluated from identical state/reset pairs. Each mechanism therefore had 24 paired rollouts and 2400 possible steps.

Table X and Fig. 12 show the resulting separation. The poisoned raw snapshots violate 23/24 rollouts and complete only one. Clean, commit, freeze, the permanent five-step backup shield, and Safe-Control-Gym’s official linear model predictive safety certification (MPSC) mechanism all have zero observed failures and complete 24/24. The shield intervenes on 511/2400 steps with no rejection, while MPSC intervenes on 2400/2400 with no infeasible solve. Median/p95 online latency is 221.925/238.216 ms for the shield and 59.657/76.055 ms for MPSC, compared with 0.026–0.027/0.028–0.029 ms for raw snapshot evaluation. These values are single-process wall-clock measurements rather than worst-case execution-time guarantees.

The safe mechanisms had different utility. Mean reward was −0.023 for freeze, −0.024 for clean, −0.056 for commit, −0.195 for the shield, and −0.457 for MPSC. Paired rollout deltas relative to freeze are respectively −0.00096 (paired-block bootstrap 95% interval [−0.00141, −0.00055]), −0.03350 ([−0.04754, −0.02182]), −0.17208 ([−0.18601, −0.15886]), and −0.43393

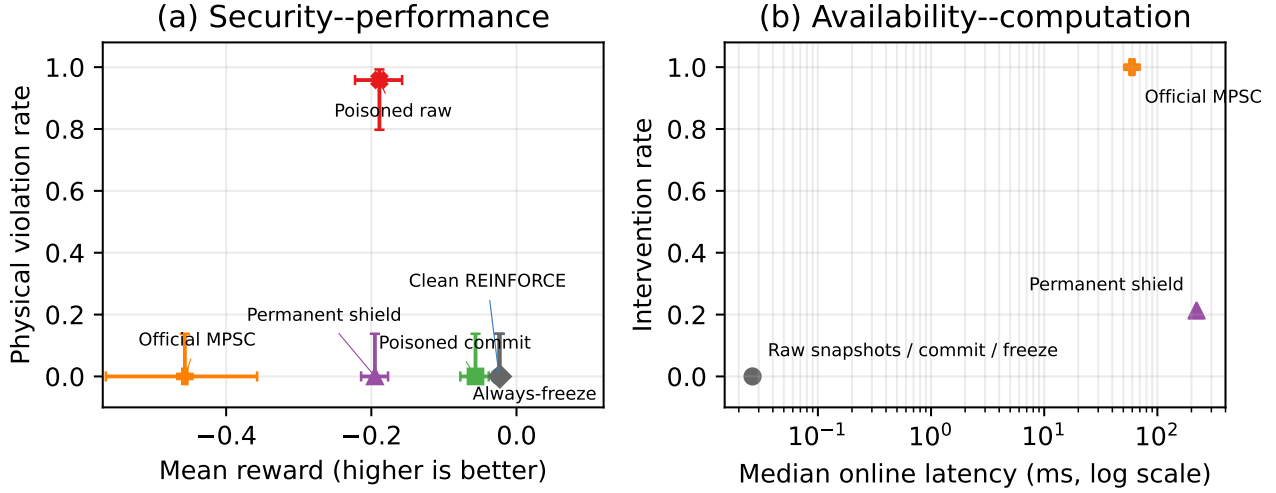


Fig. 12. The unified frontier on the same paired state blocks. Error bars in (a) are 95% Wilson intervals for violation rate and paired-block bootstrap intervals for mean rollout reward, conditional on the three locked learner snapshots. Panel (b) separates the negligible raw policy latency from the permanently-online protection cost; raw, commit, and freeze mechanisms overlap near 0.026 ms and zero intervention.

TABLE VII
COMMIT AVAILABILITY AND OFFLINE COMPUTATION ON THE
INDEPENDENT 144-STATE CANDIDATE POOL. COMMON ADMISSION
REQUIRES BOTH FREEZE AND THE COMMITTED SNAPSHOT TO PASS THE
100-STEP, 0.003-GUARD CERTIFICATE.

Seed	Common/144	Commit frac.	Offline (s)
2040	143/144 (99.3%)	0.55	8.49
2041	144/144 (100.0%)	0.40	9.32
2042	130/144 (90.3%)	0.70	8.50

($[-0.54016, -0.33763]$); none improved on freeze in any of 24 blocks. These values characterize the attack-evaluation frontier. The benign adaptation study below evaluates learning utility separately.

Table VII makes the snapshot-admission cost explicit. The common operating-set coverage is 99.3%, 100.0%, and 90.3% for the training runs indexed by 2040–2042. Reconstructing trusted-state admission and the 21-point contiguous backtracking took 8.49–9.32 s per run and exactly reproduces the locked commit fractions 0.55, 0.40, and 0.70. Offline commit therefore avoids permanent per-action computation but pays finite certificate coverage and update latency. The comparison distinguishes three contracts: certify a snapshot before raw deployment, retain a filter permanently, or stop adaptation.

M. Trusted-State-Anchor Infrastructure

A trusted state anchor shrinks certificate uncertainty but does not sanitize the reward record. The artifact varies anchor period $P \in \{1, 3, 6, 12\}$, requiring 12, 4, 2, or 1 calls over 12 updates. Every condition retains all 16 certificate centers and has 0/24 violations on new paired blocks. Periods 1–6 preserve retained fractions 0.55/0.40/0.70; at $P = 12$ they become 0.50/0.40/0.65. Neither quality grid changes fractions at 0.05 resolution, and no condition improves reward over

freeze. Full uncertainty, timing, and paired-reward rows remain in the artifact; hardware latency, energy, authentication, and monetary cost are not measured.

N. Boundary I: Official PPO

The official pretrained PPO policy is safe on 72/72 held-out rollouts, while a separately constructed malicious target is unsafe on 72/72. Under the locked $L_\infty \leq 0.5$ reward budget, the 16-batch checkpoint has 0/24 clean and 1/24 poisoned failures; at 24 batches both have 0/24. The pre-specified physical and two-checkpoint margin conditions failed, so the planned multi-run extension was not executed. The audit showed checkpoint-level update redirection without a stable deep-PPO unsafe-policy effect.

O. Boundary II: Benign Adaptation Utility

In the locked benign-utility audit under a 0.003 N differential actuator bias, freeze, clean adaptation, and LifecycleGate all have 0/12 violations and mean rewards -0.03617 , -0.04714 , and -0.08097 . LifecycleGate accepts a nonzero update in 11/12 batches (mean fraction 0.8208) but improves no paired rollout over freeze. This met the preregistered stopping criterion, so the planned three-run extension was not executed. Parameter motion and certificate acceptance did not establish useful adaptation in this setting.

P. Measurement and Validity Boundaries

The CasADi gates use fixed finite state sets; physical outcomes use independent PyBullet rollouts. They are sampled finite-horizon certificates, not continuous-state invariant-set proofs. For the benign shift, maximum action implementation error is 2.61×10^{-11} N. A corrected calibration metric left run indices, states, bias grid, and threshold unchanged. Realized state violations and pre-clip saturation are reported separately.

IX. RELATED WORK

Runtime assurance and safe controller updates. Simplex architectures retain a decision module and a trusted baseline during online upgrades [1]. Neural and barrier-based Simplex extend this model through online retraining, reverse switching, and switching conditions under bounded model error [2], [3]. Model-predictive safety certification (MPSC) and runtime shields similarly preserve recoverability or prevent unsafe actions [19]–[21]. These mechanisms certify a resident composition. Our setting instead considers a transition in which the raw policy later receives action authority.

Policy repair, safe projection, joint controller/certificate repair, and robust conformal transfer establish evidence for an updated policy before use [4]–[6], [22], [23]. Shielded transfer likewise collects independent generalization evidence before removing runtime authority [11]. These approaches re-establish evidence for the deployment object. The present work studies the complementary case in which protected adaptation is followed by raw release under adversarial update provenance.

Shield retirement and attacked state semantics. Carr et al. [9] study a partially observable bootstrap-to-retirement lifecycle and compare sudden with smooth shield removal. Their learning signals are trusted, and their study evaluates the behavioral effect of shield removal. We use this published lifecycle as a third-party case study and add bounded reward-record poisoning during the protected bootstrap. The resulting experiment concerns evidence reuse across the same authority transition, not the soundness of Carr et al.’s original claims.

Secure estimation characterizes when sensor attacks can be detected or corrected [12], [24]. Severe-attack CBF filtering constructs plausible-state sets and safe-action kernels when the state cannot be reconstructed uniquely [10]. Our formal model consumes such a plausible-state set and asks whether an adversarially updated action map remains inside its certified kernel. Related lines learn barriers, certify reinforcement-learning robustness, or solve adversarial reach-avoid problems under trusted learning signals [25]–[30].

Policy and reward poisoning. Policy poisoning has been studied in batch reinforcement learning and control [32]. Online attacks can perturb individual rewards under an ℓ_∞ budget and adapt to the learning process [33]. Extensions cover policy-gradient learners without known dynamics and black-box deep reinforcement learning [34], [35]. Targeted reward or environment poisoning can teach a chosen policy, and corrupted reward channels are a recognized reinforcement-learning hazard [36], [37].

BADControl poisons operational data in PID and LQR loops to implant a vulnerability activated by a physical trigger [31]. Our contribution concerns a different composition: bounded reward-record poisoning affects a policy trained behind runtime assurance, and failure appears after authority retirement without an activation input. Online reward-poisoning defenses model adaptive attacker-defender interaction [38]. We instead separate authenticated provenance, trusted recomputa-

TABLE VIII
CLOSEST WORK ALONG THE THREE AXES COMBINED IN THIS PAPER:
ADVERSARIAL ONLINE UPDATE DATA, REMOVAL OF
RUNTIME-ASSURANCE AUTHORITY, AND INDEPENDENT EVIDENCE FOR
THE RELEASED RAW POLICY.

Work	Adversarial update log	Authority removed	Independent raw-policy evidence
Simplex / Neural Simplex [1], [2]	no	no (monitor resident)	n/a (composition certified)
Bb-Simplex [3]	no (trusted re-training)	no	n/a
Policy repair / MPSC [4], [19]	no (trusted evidence)	repaired, verified	repaired
Hsu et al. [11]	no	yes	yes (transfer evidence)
Carr et al. [9]	no	yes (sudden/smooth)	empirical retirement
BADControl [31]	operational data, physical trigger	n/a	n/a
Controlled study	yes (reward record)	yes (raw release)	transition evidence tested
Third-party case study	yes (reward record)	yes (Carr retirement)	transition evidence tested

tion, semantics-dependent detection, commit admission, and downstream runtime containment.

Conic reward-influence analysis. Fractional programs can be homogenized into convex forms under suitable numerator and denominator structure [14]. Disciplined quasiconvex programming provides the associated fixed-level feasibility and bisection pattern [17]. Projection and feasibility over homogenization and second-order cones are classical [15], [16]. We apply these tools to the normalized image of a centered reward-to-go zonotope. Global gradient-norm clipping introduces a second norm branch, while unit-positive-numerator normalization removes the singular feasible point.

X. DISCUSSION

The evidence supports an architectural vulnerability with conditional exploitability. Protected adaptation certifies the executed composition while resident authority remains active. Permanent raw release changes the subject, authority, provenance, information, and horizon of that evidence. The third-party case study locates this transition in a published lifecycle, and the controlled study isolates the effect of adversarial reward provenance. Neither result requires the shield or filter itself to be unsound.

The attack has a structured opportunity region, but entry into that region does not determine success. Across the locked obstacle corpus, every positive outcome occurred at low retirement-state disagreement, and no high-disagreement run was positive. This pattern makes high disagreement a strong empirical exclusion condition. Low disagreement instead indicates available headroom. The fresh REINFORCE pair, seven low-band corpus runs, and the SAC battery show that headroom is insufficient for the fixed attack. Attackability therefore follows the realized retirement state more closely

than the configured seed identifier, while residual learner and trajectory dependence remains within the opportunity region.

This distinction also bounds the operational meaning of the marker. Retirement-state disagreement is observed after the protected bootstrap and therefore characterizes a realized deployment state. It is not an attacker-side trigger available before poisoning begins. Such a trigger would require evidence that earlier disagreement trajectories predict the eventual opportunity region. The reported AUC is consequently a descriptive measure over the locked corpus rather than an externally validated classifier.

The defense results separate integrity from authority. Provenance binding and trusted reward recomputation remove the evaluated upstream channel. Task-aware checks detect the effective fixed attack under additional semantic assumptions. Commit admission limits which raw snapshot receives authority, whereas resident predictive authority and permanent filtering contain unsafe policy proposals at runtime. The one-step resident kernel shows that residence alone is insufficient; the decision rule must preserve recoverability over the relevant horizon. These contracts incur different coverage, intervention, latency, and reward costs.

The formal results have bounded verification scope. The current- and future-history gates state certificate obligations, and the linear parameter gate is exact on its finite abstraction. The continuous lift requires a justified joint reachable-set cover, regularity bounds, model-error bounds, and a robust margin. The empirical coverage audit probes these premises but does not establish continuous-state invariance. The exact reward-influence result covers a fixed standardized-advantage batch with global gradient clipping. Coordinatewise parameter clipping and multi-batch attack construction remain outside that result.

The empirical generality is also bounded. The end-to-end physical effect is established for low-dimensional Monte Carlo policy-gradient learners. The learned-baseline variant shares the same return and attack channel rather than forming an independent learner family. The official Safe-Control-Gym PPO audit showed checkpoint-level redirection without stable physical failure. On the third-party lifecycle, PPO and SAC contained individual positive training realizations but did not provide reliable success across paired runs. The retirement-state marker is obstacle-scoped; the second environment also contained protective outcomes.

Finally, the threat model begins after bounded reward-ingestion access. It does not estimate the prevalence of that access across products or model service exploitation. The effective fixed attack is detectable by the tested task-aware checks, and the sign-respecting variants did not retain deployment harm. The trusted-anchor audit models bounded error only for altitude and pitch and reports calls rather than hardware-specific latency, energy, authentication, or monetary cost. These boundaries limit claims about stealth, cross-architecture prevalence, and deployment overhead without changing the authority-transition result.

XI. CONCLUSION

Protected adaptation and permanent raw release require different security evidence. Runtime assurance can keep the executed composition safe while adversarial reward provenance changes the policy that later receives action authority. Carr et al.'s shield-retirement workflow provides an independent lifecycle instance of this transition. Bootstrap-only poisoning increased raw-policy violations at retirement by factors of 3.6 and 1.6 in two of three provenance-complete paired REINFORCE training runs. The freshly regenerated third pair was unchanged, and protected execution remained violation-free.

The controlled Safe-Control-Gym study isolated the same evidence boundary. After the attack and five-step release check were fixed, poisoned permanent release failed on 27/120 snapshot-state pairs, compared with 2/120 for clean release. Resident predictive authority incurred 0/120 failures and switched before every corresponding poisoned failure. A one-step resident kernel still failed on 14/71 accepted pairs, showing that repeated recoverability checks, rather than residence alone, supplied the observed protection.

Exploitability remained conditional on the training realization. Across 50 locked obstacle pairs, all 11 positive outcomes had clean retirement-state disagreement below 0.06, and none of 32 high-disagreement runs was positive. Seven of 18 low-disagreement corpus runs were non-positive. The separate fresh REINFORCE pair was also low-disagreement and non-positive. Low disagreement therefore identifies an opportunity region without predicting attack success. Because the measurement is taken at retirement, it characterizes the realized state rather than an online attacker-side trigger.

The resulting design rule is contract-specific. Provenance binding and trusted recomputation close the upstream reward-record channel. Commit admission constrains which raw snapshot can receive authority, while resident predictive authority and permanent filtering contain downstream physical consequences. Evidence collected for an assured composition remains bound to its subject, authority, provenance, information, and horizon. Raw release must re-establish the obligations changed by the authority transition.

APPENDIX A

CERTIFICATE COVERAGE AND MODEL MISMATCH

Reporting zero violations only on certificate-admitted states can be vacuous. We therefore lock a separate CasADi-PyBullet confusion audit before execution. It reconstructs the 12 clean, poisoned, commit, and freeze snapshots from the original traces, draws 144 states from three disjoint source seeds, performs no certificate admission, and compares horizons 20/50/100 and guard margins 0/0.001/0.003/0.005/0.01. The primary configuration is the pre-existing 100-step, 0.003-guard commit configuration; the grid is characterization rather than a selector.

Table IX evaluates 1728 snapshot-state pairs. CasADi certifies 1293; none violates the PyBullet state box within 100 steps. Clean, commit, and freeze coverage is 0.986, 0.944, and

TABLE IX

CASADI-PYBULLET COVERAGE AT THE PRE-EXISTING 100-STEP, 0.003-GUARD CONFIGURATION. EACH ROW POOLS THREE LEARNER SNAPSHOTS OVER THE SAME 144 STATES.

Snapshot	Pairs	Physical viol.	Certified	False accept	False reject	Coverage
Clean REINFORCE	432	6	426	0	0	0.986
Poisoned action-only	432	403	27	0	2	0.062
Poisoned commit	432	21	408	0	3	0.944
Always-freeze	432	0	432	0	0	1.000

1.000. The raw poisoned snapshots physically violate 403/432 pairs and have certificate coverage 0.062, so their low coverage is informative rejection rather than vacuous evidence for a safe deployed snapshot. Across all mechanisms the certificate conservatively rejects five physically safe pairs. Maximum action-interface error and actuator saturation are zero.

The grid exposes a model-mismatch boundary. Without a guard, poisoned snapshots produce 1, 2, and 1 false acceptances at horizons 20, 50, and 100. Every locked positive guard removes observed false acceptance, with increasing false rejection as the guard grows. The value 0.003 was the declared primary configuration before this audit. Zero observed false acceptance on the finite evaluation distribution does not establish continuous-state invariance.

APPENDIX B

UNIFIED PAIRED SECURITY-AVAILABILITY FRONTIER

ETHICS CONSIDERATIONS

This paper studies an attack on safety-critical learning-enabled control. All experiments run in simulation using Safe-Control-Gym with PyBullet dynamics. No physical plant, vehicle, deployed industrial system, human subject, or personal data was involved. Reported failures are simulated constraint violations used to compare deployment contracts.

The evaluated threat begins *after* compromise of a reward-ingestion principal in the asynchronous learning-data plane and grants only bounded scalar writes to reward records. Initial access, product-specific exploitation, and evasion of log-integrity controls fall outside the artifact and evaluation. The experiments characterize the consequence of a least-privilege data-integrity gap for evidence inheritance at raw-policy release.

The study evaluates countermeasures at both sides of the authority transition. Origin-bound reward records and trusted recomputation remove the upstream precondition. Resident predictive authority, commit admission, and permanent filtering contain the downstream physical consequence. Simple task-aware reward checks also detected the effective fixed attack. Carr et al.’s published workflow supplies a third-party lifecycle case study; no fielded vendor or product is evaluated.

OPEN SCIENCE

We support the NDSS open-science policy. The artifact includes row-level result files, exact commands and run namespaces, upstream revisions, model files, dependency versions, source-linked tables, and SHA-256 checksums for locked files.

Automated checks audit table provenance, claim locks, and format. We intend to release the code and data required to reproduce the results under an open license upon publication. The artifact retains the anchor, coverage, timing, PPO, utility, and protocol-chronology boundaries.

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TABLE X

UNIFIED PAIRED SECURITY–AVAILABILITY FRONTIER. WILSON 95% INTERVALS ARE $[0, 0.138]$ FOR EVERY 0/24 VIOLATION COUNT AND $[0.798, 0.993]$ FOR POISONED RAW. INTERVENTION IS A FRACTION OF EXECUTED STEPS. † DENOTES REWARD TRUNCATED AT THE FIRST SAFETY FAILURE AND THEREFORE NOT A UTILITY COMPARISON.

Mechanism	Viol./rollouts	Complete	Mean reward	Interv.	Latency med./p95 (ms)
Clean REINFORCE	0/24	24/24	−0.024	0.0%	0.027/0.029
Poisoned raw	23/24	1/24	−0.189†	0.0%	0.027/0.029
Poisoned commit	0/24	24/24	−0.056	0.0%	0.026/0.029
Always-freeze	0/24	24/24	−0.023	0.0%	0.026/0.028
Permanent shield	0/24	24/24	−0.195	21.3%	221.925/238.216
Official MPSC	0/24	24/24	−0.457	100.0%	59.657/76.055

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